Respiratory Modulation of Peripersonal Space Representation

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Abstract

Peripersonal space (PPS) is often inferred from tactile reaction time during an approaching sound. We tested whether instructed respiratory phase changed this SOA-dependent response differently in breath-coupled virtual reality (Viscereality) and Control. The released cohort comprised 42 participant records and 8,916 retained audio–tactile hit trials. The planned Condition × Phase × SOA mixed-model interaction was not detected, $\chi^2(1) = 0.024$, $p = .878$; categorical-SOA and partial-session-exclusion sensitivities agreed. Audio–tactile responses were 18.8 ms faster than tactile-only Baseline responses on average, but equal nominal delay labels do not establish matched cue times. None of 168 participant-by-condition-by-phase trial-level sigmoid fits met the planned $R^2 \geq .70$ threshold. The maximum saturated five-SOA trial-level R^2 ceiling was 0.430, precluding this threshold for any curve of SOA alone. Participant-boundary and boundary-related questionnaire inference therefore remained unestimated. A separate post hoc comparison of 800 uniformly processed sigmoid and linear pipelines produced four explicit shape-constrained approximations. Their visible S shapes were imposed; nested participant prediction did not improve on a constant, and unsmoothed curvature tests did not establish nonlinearity. Midpoint and

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steepness estimates remain assumption-dependent descriptions. An exploratory paired ordinal comparison in 41 participants found higher post-intervention spatial-frame codes under Viscereality (exact sign-test $p = 0.0037$). Pre-experiment MAIA-2 Trusting correlated with the numerical Viscereality–Control difference ($\rho = 0.376$, unadjusted $p = 0.0160$); its moderation estimate remained positive with order and HC3 sensitivities. These post hoc associations are not evidence of mediation. A further exploratory extension produced 168 individual sigmoid proxies and common-plateau displays, but midpoint reliability was low and all main refit-bootstrap association intervals included zero. An audit found two identical behavioural recordings with reversed intervention labels and different questionnaire responses; full-release behavioural inference is provisional pending source mapping. The full data-supported pipeline, diagnostics, and downloadable results run in a marimo browser companion. These data do not demonstrate the predicted Viscereality-specific respiratory modulation; confidence intervals and source-timing and respiratory-compliance limits constrain interpretation.

Keywords: peripersonal space, respiration, virtual reality, interoception, embodied cognition

Analysis draft for author review. Results from 42 participant records are reproduced by the browser companion. Two records share an identical behavioural recording with reversed intervention labels; source mapping must be resolved before behavioural inference can be finalized. Collection-history and ethics-statement details also require author review before submission.

1 Introduction

Embodied-cognition accounts treat the mind as built on, and shaped by, the sensorimotor exchanges between a body and its environment (Barsalou, 2008a, 2008b; Glenberg, 2010). One component of that exchange is *peripersonal space* (PPS), the multisensory region immediately surrounding the body within which incoming signals are integrated and prepared for action (di Pellegrino & Làdavas, 2015; Noel, Bertoni, & Serino, 2021; Rizzolatti, Fadiga, Fogassi, & Gallese, 1997; Serino, 2019). Audio–tactile and visuo–tactile paradigms measure tactile responses at different points along an approaching or receding stimulus trajectory (Canzoneri, Magosso, & Serino, 2012; Teneggi, Canzoneri, di Pellegrino, & Serino, 2013). The resulting response gradient is often fitted with a sigmoid; when its shape and parameters are supported by the data, the midpoint and slope can describe the transition's location and sharpness (Basile et al., 2024; Bufacchi & Iannetti, 2018; Serino, 2019). Auditory–tactile

reaction-time gradients can also reflect temporal expectancy and other design features, so a fitted curve alone does not establish a spatial PPS boundary (Holmes, Martin, Mitchell, Noorani, & Thorne, 2020). On current accounts, PPS is not a distinct cortical module but a case of multisensory integration operating under body-centred coordinate frames with an additional proximity-gating term (Bertoni, Chauhan, Noel, & Serino, 2025; Noel, Blanke, & Serino, 2018; Straka, Noel, & Hoffmann, 2022), and the same electrophysiological signature has been linked to self-referenced consciousness states in sleep and in disorders of consciousness (Bertoni et al., 2026). A well-supported response curve can therefore offer a behavioural estimate of a transition along the probed trajectory. The remainder of this introduction traces a single argument in five steps. First, plasticity findings indicate that behavioural PPS estimates can follow the brain's body estimate rather than the physical body (Noel et al., 2021; Serino, 2019). Second, interoceptive signals belong in that estimate, yet the channel by which they reach it is unspecified (Ardizzi & Ferri, 2018; Bogdanova et al., 2021; Salvato, Richter, Sedeño, Bottini, & Paulesu, 2020). Third, respiration is a plausible channel (Allen, Varga, & Heck, 2021; Saltafossi et al., 2025). Fourth, the direct empirical test and the corresponding plasticity manipulation are absent from the published literature (Karakoc, Lucifora, Massimino, Nucera, & Vicario, 2025; Spaccasassi, Romano, & Maravita, 2019). Finally, the present study tests both.

1.1 Peripersonal space is plastic and tracks the state of the body

The PPS sigmoid is malleable (Bufacchi & Iannetti, 2018; Karakoc et al., 2025; Noel et al., 2021; Serino, 2019). Tool use extends it to incorporate the tool (Iriki, Tanaka, & Iwamura, 1996; Maravita & Iriki, 2004); cooperative social interaction merges self and partner space (Teneggi et al., 2013); body-swap and full-body illusions relocate PPS with the illusorily perceived self-location (Noel, Grivaz, et al., 2015; Noel, Pfeiffer, Blanke, & Serino, 2015); and vestibular stimulation reshapes it in line with altered spatial orientation (Pfeiffer, Noel, Serino, & Blanke, 2018). A systematic review of immersive-VR manipulations of PPS reports a comparable pattern. Across tool-use, social, and embodiment paradigms, the boundary tracks the multisensorially constructed body estimate that the VR display engages rather than the physical body (Karakoc et al., 2025). Read together, these manipulations are consistent with the view that PPS is anchored to the multisensorially estimated state of the body rather than to the physical body per se (Blanke, Slater, & Serino, 2015; Grivaz, Blanke, & Serino, 2017; Noel et al., 2021; Serino et al., 2015). If PPS follows the brain's ongoing body estimate, then any signal that feeds that estimate, including afferents arising from inside the body, should in principle also shape PPS. The next subsection turns to evidence

that interoceptive signals do.

1.2 Interoception and the softening of the self–world boundary

A more recent and smaller literature extends this logic from exteroceptive to interoceptive signals. Ardizzi and Ferri (2018) reported that individuals with higher cardiac interoceptive accuracy have *narrower* PPS boundaries, with the effect moderated by a dispositional tendency to attend to internal sensations. Brief focused-attention meditation has been reported to reduce the sharpness of the PPS sigmoid without contracting its extent (Chiarella, De Pastina, Raffone, & Simione, 2024); guided meditation has been reported to abolish the near–far attentional enhancement of auditory evoked responses in non-meditators (Nguyen, Pipingas, Stough, & Downey, 2020); mindfulness training has been associated with looser perceived body boundaries and a shift toward allocentric spatial frames (Hanley, Dambrun, & Garland, 2020); and sensory-attenuation contexts such as Floatation-REST have been associated with body-boundary dissolution that mediates anxiety reduction (Hruby et al., 2024). A kernel-density meta-analysis of neuroimaging studies locates the anatomical convergence between interoceptive and exteroceptive bodily-self processing in bilateral supramarginal gyrus and right-lateralised peri-central cortex, that is, in PPS-relevant parietal territory (Salvato et al., 2020). Considered together with the broader literature on breath, heart, and gut as contributors to self-experience (Barrett, 2016; Bogdanova et al., 2021; Park & Blanke, 2019; Seth, 2013), these findings are consistent with treating PPS as an *intero–exteroceptive interface*, a zone in which interoceptive priors about the body meet the statistics of the surrounding world (Bogdanova et al., 2021; Noel et al., 2021; Salvato et al., 2020). What remains underspecified is *which* interoceptive signals reach the PPS computation, and through which temporal channel.

1.3 Respiration as an organiser of perception and multisensory integration

Among interoceptive channels, breath is one practical candidate. Unlike other autonomic rhythms, respiration is both voluntarily controllable and automatically driven, and it carries chemoreceptor, mechanoreceptor, and proprioceptive afferents that converge on brainstem, insular, and limbic targets (Chan, Lu, & Davenport, 2024; Heck et al., 2017; Varga & Heck, 2017; Zelano, Jiang, & Gottfried, 2022). Evidence accumulated over the past decade indicates that respiratory phase modulates cortical excitability and perception beyond the olfactory system. Nasal inhalation has been associated with entrained oscillations in

piriform cortex, amygdala, and hippocampus, and with speeded fearful-face discrimination and episodic retrieval (Mizuhara et al., 2023; Zelano et al., 2016). Respiration has been linked to posterior alpha power and to visual detection sensitivity, with excitability changes preceding performance fluctuations (Kluger & Gross, 2021). Tactile detection has been reported to peak in the first quadrant after expiration onset, and participants phase-lock their breathing to expected stimuli (Grund et al., 2021). Breath–heart coupling via respiratory sinus arrhythmia has been related to heartbeat-evoked potentials and to the gating of somatosensory input (Heck et al., 2022; Zaccaro, Perrucci, Parrotta, Costantini, & Ferri, 2022). Across modalities, then, respiration appears to contribute to the temporal organisation of sensory processing rather than acting as a passive background state (Heck et al., 2022; Zelano et al., 2022).

Of direct relevance to the present work, Saltafossi et al. (2025) used the race-model inequality to provide evidence that respiratory phase modulates multisensory integration. Audio–tactile and audio–visual integration peaked at the expiration-to-inspiration transition, with multisensory gain dissociating from unimodal speed. Predictive-coding accounts formulate this by casting respiration as a temporal scaffold along which the brain allocates precision to competing sources of evidence about the body and the surrounding world, with breath phase modulating local and global neural gain across interoceptive and exteroceptive channels (Allen et al., 2021; Brændholt et al., 2023; Engelen, Solcà, & Tallon-Baudry, 2023). Because PPS can be characterised as multisensory integration operating under body-centred reference frames with an additional proximity-gating term (Bertoni et al., 2025; Noel et al., 2021, 2018), the respiratory-phase organisation that has been observed for generic multisensory integration may, in principle, also be observable in the PPS sigmoid. The next subsection sets out why this prediction has not yet been tested, and why a second, paradigmatic gap accompanies it.

1.4 A missing test and a missing manipulation

Two literatures have approached the same question without directly addressing it. PPS is a body-referenced multisensory integration process that has been shown to be shaped by interoceptive signals (Ardizzi & Ferri, 2018; Bogdanova et al., 2021; Salvato et al., 2020), and respiratory phase has been reported to modulate perception and multisensory integration more generally (Grund et al., 2021; Kluger & Gross, 2021; Saltafossi et al., 2025). To our knowledge, no published study has delivered PPS-task stimuli at controlled respiratory phases and fit the sigmoid separately under each, and recent reviews of PPS computational models and of immersive-VR PPS paradigms describe no such design (Bertoni et al., 2025;

Karakoc et al., 2025). The one empirical study that embeds a breathing manipulation within a PPS paradigm used breathing as an anxiety-induction procedure rather than as a phase manipulation (Spaccasassi et al., 2019), and the "embreathment" illusion shows that breathing feedback on a seen avatar can shift corporeal awareness without engaging the PPS sigmoid directly (Allard et al., 2017; Betka, Adler, Similowski, & Blanke, 2022; Betka et al., 2020; Monti, Porciello, Tieri, & Aglioti, 2020). The gap is therefore at the intersection of the two literatures. Respiratory phase has been shown to modulate generic multisensory integration, but not the body-referenced multisensory integration that PPS isolates.

Addressing this first gap invites a second, paradigmatic question. The plasticity manipulations that have informed PPS theory, including tool use, rubber-hand and full-body illusions, and audio–tactile far-and-near pairing, share the structural feature of introducing a synchronous multimodal signal that causal-inference accounts treat as evidence of a shared generative source (Chancel, Ehrsson, & Ma, 2021; Costantini et al., 2016; Parise & Ernst, 2016; Samad, Chung, & Shams, 2015). To our knowledge, no analogous manipulation has been reported for a purely visceral–exteroceptive pairing, and the systematic review of immersive-VR PPS paradigms catalogues tool-use, social, and embodiment variants without listing a bioresponsive physiological-coupling manipulation (Karakoc et al., 2025). The nearest precedent is cardiovisual. Sel, Azevedo, and Tsakiris (2016) reported that a pulsing visual shade locked to the observer's heartbeat increased identification with an otherwise ambiguous self–other face and modulated heartbeat-evoked potentials, with the effect scaling with individual interoceptive accuracy. Bioresponsive virtual reality renders the corresponding experiment for the respiratory channel technically feasible. Visual elements of an immersive environment can be tied in real time to the participant's ongoing respiratory signal, an arrangement recently described as *exteroceptive–interoceptive sensory substitution* (Goral et al., 2024). In the taxonomy of PPS-plasticity paradigms this corresponds to a bioresponsive environmental-coupling manipulation, and, to our knowledge, it has not yet been reported in the empirical record (Belli, Felisatti, & Fischer, 2021; Belli & Fischer, 2024; Karakoc et al., 2025; Serino et al., 2017). The present study addresses both gaps in a single design.

1.5 The present study

Participants perform an audio–tactile PPS task (Canzoneri et al., 2012; Teneggi et al., 2013) under instructed 4–4–4–4 box breathing, cycling through inhalation, post-inhalation hold, exhalation, and post-exhalation hold. Audio–tactile stimuli are scheduled during the two breath-hold phases, permitting separate SOA-response curves for post-inhalation and

post-exhalation holds under the assumption of compliant breathing. Participants complete this task under two VR conditions with both orders represented in the pooled cohort. In the *Viscereality* condition, an ovular orb in front of the participant expands and contracts in real time with their breathing, providing a visual environment whose spatial dynamics are coupled to the breath signal (Goral et al., 2024). In the *Control* condition, the same task is performed against a dark field containing only the central breath-performance indicator that scales with respiratory excursion, providing minimal visual feedback for breath performance without spatial coupling.

The preregistered main prediction was that, if coupling an exteroceptive spatial signal to the interoceptive breath signal engages processes linked to body ownership and PPS plasticity (Ardizzi & Ferri, 2018; Chancel et al., 2021; Samad et al., 2015), the breath-coupled condition would show a larger PPS response during post-inhalation than post-exhalation holds, yielding a Condition × Phase × SOA interaction. A secondary Control-only Phase × SOA test asked whether scheduled respiratory phase modulated the SOA response without spatial visual coupling (Grund et al., 2021; Kluger & Gross, 2021; Saltafossi et al., 2025). The registered parameter predictions concerned the sigmoid midpoint interaction and a less sharp boundary under Viscereality, conditional on interpretable sigmoid fits (Chiarella et al., 2024; Hanley et al., 2020; Hruby et al., 2024; Nguyen et al., 2020). Trait interoceptive awareness (MAIA-2) and pictographic measures of perceived body boundaries, spatial frame of reference, and small-self experience were exploratory correlates rather than additional confirmed tests.

2 Materials and Methods

2.1 Apparatus

Auditory and tactile stimuli were delivered through a Komplete Audio 6 MK2 USB audio interface (Native Instruments GmbH, 2020). The audio channel was routed to a pair of Sennheiser HD 560S open-back reference headphones (Sennheiser electronic GmbH & Co. KG, 2020), worn over the ears, and the tactile channel was routed to a Woojer Strap 4 haptic transducer (Woojer Ltd., 2024), worn over the chest just above the breast area. Placement was standardised across participants by following the positioning illustration printed on the strap itself, which the operator used as a fitting guide. Participant responses were collected with a standard USB mouse. Respiratory compliance was monitored throughout the session with a chest-mounted respiratory belt transducer positioned around the lower ribcage. In the Viscereality condition, participants additionally wore a head-mounted display delivering

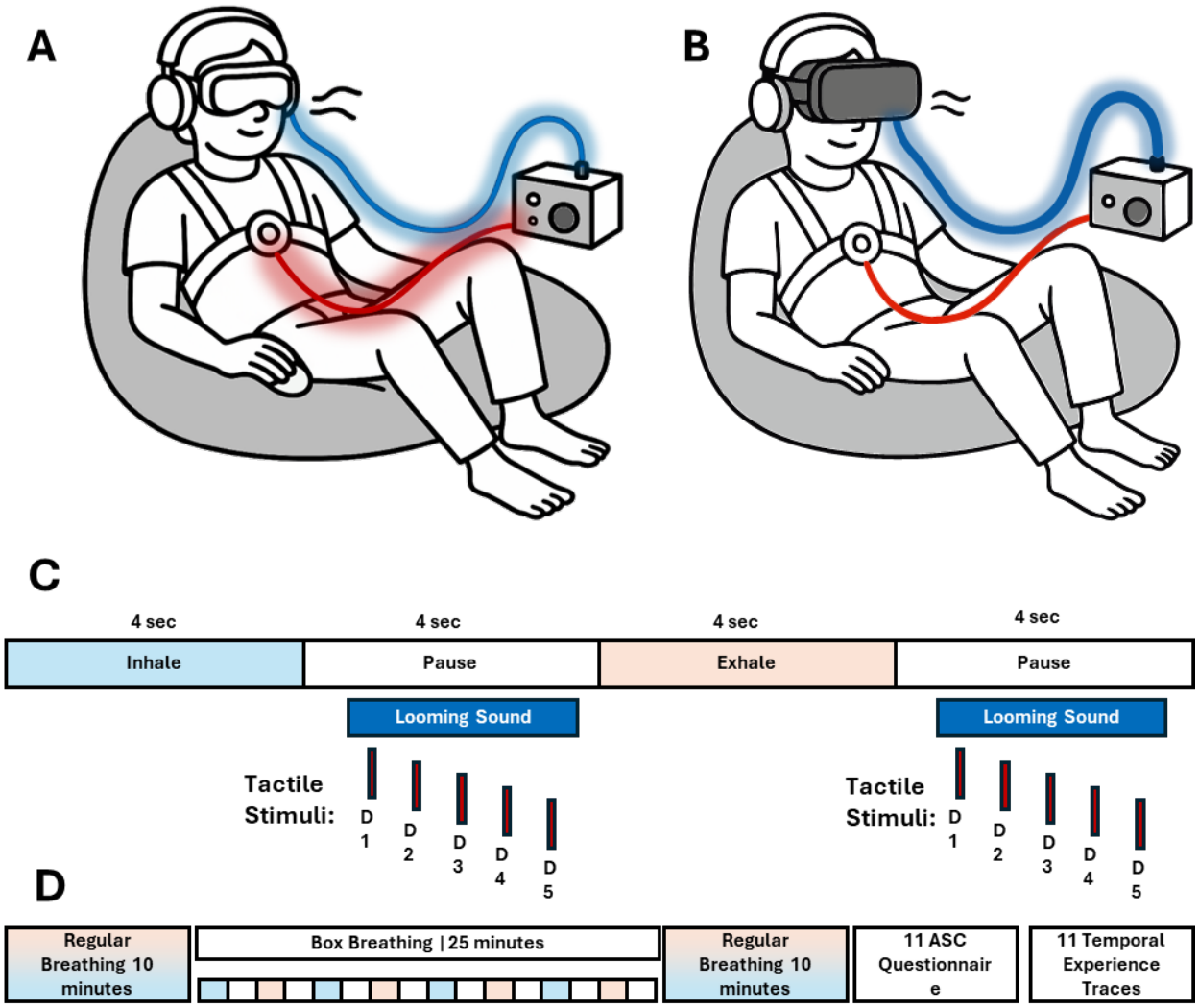


Fig. 1 Experimental setup and protocol. A, Control condition setup with participant wearing headphones (blue cord) and a chest-mounted vibrotactile stimulator (red cord) positioned above the breast area as depicted on the strap itself, providing only the minimal breath-performance feedback indicator without spatial cues. **B**, Viscereality condition setup with the breath-responsive VR headset providing visual feedback of respiratory state. **C**, Box-breathing protocol with looming sounds and tactile stimuli presented during breath-hold phases. D1–D5 represent stimulus onset asynchronies between audio and tactile stimuli.

the breath-coupled visual environment. In the Control condition, participants viewed the same dark field used as the black-screen control in Study 3, containing only the central breathing performance indicator that scaled with respiratory excursion. This provided minimal feedback stimuli for breath performance without the spatial depth cues, color, or aesthetic properties of the Viscereality environment, and participants did not wear an eye shade (Fig. 1).

2.2 Stimulus generation

The box-breathing guide was built from two pre-recorded human-voice audio files, one for the inhalation half-cycle and one for the exhalation half-cycle (Robert, 2022). Each file voiced a count of "two, three, four, hold" over four seconds so that a full 4–4–4–4 cycle of inhalation, post-inhalation hold, exhalation, and post-exhalation hold lasted sixteen seconds. Files were loaded at their source rate and resampled to 44.1 kHz using `librosa` where available and otherwise `scipy.signal.resample` (McFee et al., 2015; Virtanen et al., 2020). Amplitude was attenuated to 30% of source level so that the voice guide remained audible without masking the looming stimuli. The guide occupied the right channel of the per-trial stereo WAV, with the left channel set to silence over the same four-second span.

Binaural looming stimuli were initially synthesised from four noise types, namely pink, white, brown, and blue noise; later acquisition versions used a subset, predominantly pink and white (Brinkmann, Lindau, Weinzierl, Geissler, et al., 2017). For each noise type, a 3-second trajectory was rendered in which a point source approached the head frontally from 1.1 m to 0.1 m at a constant 33.3 cm/s, at zero azimuth and zero elevation. Source-to-ear distances were computed for an 18 cm interaural distance, and amplitude was scaled by the inverse square of source-to-ear distance. Head-related transfer function filtering used the measured FABIAN head-related impulse response at neutral head-above-torso orientation, specifically the `FABIAN_HRIR_measured_HATO_0.sofa` file from the FABIAN database hosted on the Technische Universität Berlin DepositOnce repository (Brinkmann, Lindau, Weinzierl, Geissler, et al., 2017; Brinkmann, Lindau, Weinzierl, van de Par, et al., 2017). The SOFA file was parsed with the `sofar` Python library (Berzborn, Ackermann, Klein, & Brinkmann, 2023), the left and right impulse responses for the azimuth-zero, elevation-zero position were extracted, and each was convolved with the per-channel noise trajectory using `scipy.signal.fftconvolve` (Virtanen et al., 2020). Each looming file was normalised to 95% of maximum amplitude to prevent digital clipping and was padded with 500 ms of silence before and after the 3-second approach, yielding a 4-second stimulus segment.

Tactile pulses were generated for the Woojer Strap 4 transducer, whose resonance is near 50 Hz. Each pulse was a 100 ms pure-tone 50 Hz sine at digital amplitude 1.0 with an instant-on, instant-off block envelope, chosen to deliver the maximum mechanical impulse the transducer supports (Woojer Ltd., 2024). The stimulus-generation scripts can instantiate pulses at 0, 300, 800, 1500, 2200, and 2700 ms after looming onset, with the 0 ms file serving as an audio–tactile simultaneity stimulus. The participant master blocks used the five non-zero PPS SOAs, namely 300, 800, 1500, 2200, and 2700 ms, so that tactile cues sampled the approaching sound after it had begun moving toward the body.

Each trial was compiled as an 8-second stereo WAV file at 44.1 kHz in 32-bit float using `soundfile` (Geier, 2024). The file contained a 4-second breathing-guide segment followed by a 4-second stimulus segment. Within the stimulus segment, the looming noise occupied the right channel and the tactile pulse occupied the left channel, positioned at the trial-specific SOA within the 4-second window. Baseline trials carried the tactile pulse without a looming sound, and catch trials carried the looming sound without a tactile pulse. Total trial duration was verified at assembly to within one millisecond of the 352,800-sample target.

Source-timing inspection revealed a qualification to the nominal cue schedule. In 167 hash-verified original block recordings from 14 cleared participants, tactile pulses were matched to the archived played-order CSVs; all block-median Audio–Tactile tactile cues occurred 500 ms later within the eight-second trial than Baseline cues with the same nominal SOA label. An independently decoded session from a fifteenth cleared participant showed the same block-median offset in all 12 comparable blocks. The archived generator loads looming inputs already padded by 500 ms and has a path that prepends another 500 ms to assembled Audio–Tactile trials while Baseline trials receive one prepend. This shifts absolute cue time while preserving the within-trial auditory-to-tactile SOA, verified directly where both onsets were decoded in the independent session. The other 25 cleared participants lack comparable cue-time verification, so shared nominal labels cannot establish matched absolute timing across trial types in the full cohort.

Per-participant trial sequences were generated under a three-level randomisation logic intended to preserve the planned design at the session level while limiting trial-to-trial predictability within a block. At the macro level, the Condition × respiratory Phase × five-SOA grid for Audio–Tactile trials and the trial-type counts were fixed across full sessions. Available noise types varied by acquisition version, so sound-type balance was version-specific rather than common to every participant. Within each block, trial order was drawn from a Gellermann-inspired constrained randomisation in which post-inhalation and post-exhalation trials were interleaved, with the algorithm rejecting any sequence that placed more than three consecutive trials of the same type or distributed a trial type asymmetrically across the two halves of a block (Gellermann, 1933). Each session was organised as two parts of six blocks each (12 blocks in total), with the six blocks within each part labelled a through f, and each participant was assigned one block-order permutation drawn from the 720 orderings of the six block labels, applied within both parts. This block-order randomisation distributes any block-level position effects across participants without disturbing the within-block constraints. Each block was concatenated into a single continuous WAV of the stimulus-delivery length, accompanied by a trial-order CSV that recorded the trial type, SOA, noise type, and respiratory phase per trial.

A residual source of predictability is unavoidable in this design. The 4–4–4–4 box-breathing rhythm fixes the within-cycle position of the breath-hold phases at which stimuli can be delivered, so participants can in principle anticipate *when*, within each respiratory cycle, an audio–tactile event might arrive. Removing this regularity is not feasible, because a stable respiratory cycle is the precondition for systematically assessing inhale versus exhale as a test variable; an irregular cycle would compromise phase assignment, which is the central manipulation of the study. We therefore adopted the regular 4–4–4–4 rhythm and limited predictability by other means. *Which* stimulus arrives was randomised across the noise types available in each acquisition version. *Whether* a tactile pulse arrives at all was randomised through the inclusion of catch trials (sound-only, no vibration), which required response inhibition. Together with the within-block constrained randomisation and the per-participant block-order permutation, these manipulations limited cues that a participant could exploit to anticipate any individual trial.

2.3 Trial design

The experiment used a within-subjects design adapted from the protocol used to test PPS modulation by vestibular stimulation (Pfeiffer et al., 2018), modified for scheduled audio–tactile measurement during box breathing. Each full session planned 408 trials, organised into two parts of six blocks each (12 blocks of 34 trials), with 240 Audio–Tactile trials, 120 Baseline trials, and 48 Catch trials. Audio–Tactile trials delivered a looming sound followed by a tactile pulse at one of five SOAs (300, 800, 1500, 2200, and 2700 ms; Lee et al. 2021; Mine and Yokosawa 2022; Teneggi et al. 2013). The initial acquisition version used blue, brown, pink, and white spectral noise; later versions predominantly used pink and white, the two sound types present across the whole publication cohort. Within each version, Audio–Tactile trials were planned across the 2 (respiratory phase) $\times$ 5 (SOA) design grid at 24 trials per phase $\times$ SOA cell. Baseline trials delivered only the tactile pulse, used the same five nominal delay labels, and were balanced across respiratory phase (60 inhale, 60 exhale). They provide a tactile-only response reference but have no auditory distance; their absolute cue timing cannot be presumed equal to Audio–Tactile trials at the same label. Catch trials delivered only the looming sound and required response inhibition (48 planned trials), yielding a planned catch rate of approximately 12%. Each trial lasted 8 seconds, so a full 408-trial session took approximately 54 minutes of stimulus time excluding block-level breaks and instructions (Fig. 2). One included session ended before all 408 planned trials had been observed; its missing slots and an exclusion sensitivity are documented in the Results.



Fig. 2 Planned per-session trial distribution across conditions, respiratory phases, and stimulus onset asynchronies. A full session plans 408 trials, organised as two parts of six blocks (12 blocks of 34). Audio–Tactile trials are distributed evenly across five SOAs (D1–D5; 24 trials per phase × SOA cell). Available noise types differ by acquisition version. Baseline trials have no auditory SOA or distance; they carry the same five nominal delay labels and are balanced across respiratory phase. Catch trials are balanced across phase and distributed across SOAs and the available noise types.

2.4 Stimulus delivery and response capture

At run time, each per-block WAV was fully pre-loaded into memory and streamed to the Komplete Audio 6 MK2 on WASAPI via the `sounddevice` Python binding to the PortAudio library (Geier, 2024), with a 4096-sample callback block size and a target output latency of 100 ms for stable long-block playback. Python swapped the two channels at the callback boundary so that the left channel of the file was sent to physical Output 2, wired to the Woojer Strap 4 haptic transducer, and the right channel of the file was sent to physical Output 1, wired to the Sennheiser HD 560S headphones. The audio and tactile streams originated from the same file and passed through the same stereo callback, preserving their sample alignment in the digital output stream. Relative acoustic and mechanical onset latencies at the participant were not independently measured.

Mouse clicks were monitored with the `pynput` library (Palm, 2023), and each click triggered a 1200 Hz, 50 ms synthetic click tone on a separate low-latency `sounddevice` stream that was routed only to Output 2. The click tone served two purposes. It provided the participant with immediate auditory feedback that the response had registered, and it inserted a detectable event onto the tactile channel that was captured in the loopback recording described below and used for post-hoc reaction-time estimation.

Before participant playback, the selected stereo output profile was required to pass a routing preflight in which calibration pulses were sent separately to Output 1 and Output 2 and verified through the corresponding WASAPI loopback endpoint. The documented acquisition profile used the Komplete Audio 6 MK2, with Output 1 assigned to the headphones and Output 2 assigned to the tactile transducer, but the same software could be run on another wired two-channel interface only after the same channel-separation preflight passed. Once participant playback began, the acquisition software was designed to mirror both physical outputs to a WASAPI loopback device and capture them through the `PyAudioWPatch` extension of the PortAudio interface (Smirnov, 2023). These source recordings are not presently linked and independently checked for every dataset in the publication pool. Available session WAV files and behavioural CSVs were written to participant-specific data folders with identifying metadata; only coded participant IDs appear in analysis outputs.

2.5 Reaction-time reconstruction from loopback recordings

For the loopback data to which the waveform-decoding pipeline was applied, reaction times were reconstructed in Python using `numpy` and `scipy` (Virtanen et al., 2020). In that pipeline, audio and tactile channels were identified by the ratio of mean absolute amplitude to root-mean-square amplitude, with the sparser channel assigned as tactile. Tactile cue events were then detected by smoothing the absolute tactile signal with a 10 ms rectangular envelope and thresholding at 0.45 of the envelope peak, subject to a minimum run of 60 ms and a minimum peak amplitude. The resulting cue-onset train was fitted to an 8-second trial grid by jointly minimising the median absolute error between detected onsets and the nominal SOA set, yielding a per-trial cue-onset time in the sample domain of the recorded WAV.

For each trial processed by that pipeline, the candidate looming templates for its acquisition version were trimmed to their first 1.5 s of non-silent content, downsampled fivefold for speed, and z-standardised. Each template was cross-correlated with the audio channel of the recording inside the 4.45 s to 7.85 s window relative to the trial origin, using frequency-domain convolution (Virtanen et al., 2020). The highest-scoring template above threshold assigned the trial its observed noise type and returned the template-onset timestamp, giving an independent anchor for SOA verification where such recording evidence was available.

Where the loopback recording and both event markers were available, the 1200 Hz, 50 ms click tone template was z-standardised and cross-correlated with the tactile channel starting

at each detected cue onset. Click events were accepted when the local correlation peak exceeded a dynamic threshold of eight median absolute deviations above the local median and were separated from any prior click by at least 100 ms. For such recordings, reaction time was the sample-domain difference between click-tone onset and tactile-cue onset, converted to milliseconds at 44,100 Hz. The pooled publication dataset also contains source-recorded reaction times from acquisition versions for which this exact waveform-based definition has not been independently re-harmonised. Consequently the sample-domain timing advantage cannot be assumed for every pooled trial; source-specific timing definitions and verification status are retained in the private provenance crosswalk.

An earlier 40-person source audit catalogued original-recording origins for 31 participants and verified at least one asset hash for each of 17 early-version sessions. That historical audit documented RT-onset variants and unavailable original assets; it is not a new audit of the released 42-person cohort. Source dates, waveform verification status, and acquisition-version identities are deliberately absent from the public tables. The browser companion therefore reports the limits of source-recorded RTs without reconstructing private provenance from arbitrary participant IDs. A file hash establishes source identity, not waveform remeasurement or respiratory-phase accuracy.

Trials were labelled against their scheduled trial order and, where available, checked against runtime and recording evidence. Audio–Tactile and Baseline trials with a detected response were labelled hits; Catch trials with a response were false alarms. Unobserved planned slots were kept explicit in the participant audit tables and excluded from reaction-time analysis. Hit-labelled trials without a recorded reaction time contributed to response-quality summaries but not to reaction-time models; missing reaction times were not imputed. The publication-pool analysis table contains only observed planned trials, with trial type, SOA, noise type, scheduled respiratory phase, condition, response outcome, and source-recorded reaction time. Event-marker availability and source location are recorded separately; the analysis table does not imply that every reaction time was newly decoded from a common waveform.

2.6 Peripersonal-space quantification

Peripersonal space was operationalised as SOA-dependent audio–tactile reaction time in a looming-sound tactile-detection paradigm (Canzoneri et al., 2012; Noel, Pfeiffer, et al., 2015; Teneggi et al., 2013). The tactile pulse occurred at one of five nominal SOAs (300, 800, 1500, 2200, or 2700 ms); longer SOAs corresponded to nearer nominal source positions. The registration labels these SOAs as approximately 100, 73, 50, 27, and 10 cm, although

its stated linear 110-to-10 cm, three-second trajectory instead implies approximately 100, 83, 60, 37, and 20 cm at those times. The primary predictor was therefore nominal SOA in milliseconds, not an inferred physical distance. The browser sigmoid implementation retains the registered distance labels for its viability gate; a sensitivity repeated that gate using trajectory-implied positions. It subtracts mean tactile-only Baseline reaction time within each participant × Condition × Phase cell: $RT_{AT} - RT_{baseline}$, with negative values indicating faster Audio–Tactile responses. Baseline trials have no auditory stimulus or acoustic distance. We also compared Audio–Tactile and Baseline RTs at the same nominal SOA label descriptively, without treating the labels as proof of equal cue time. Misses, Catch outcomes, and unobserved planned slots were retained for quality summaries but did not supply reaction times to these curves.

2.7 Questionnaire scoring and exploratory analysis

The released participant table contains one set of 37 MAIA-2 items and paired Control/Viscereality spatial-frame-of-reference ratings. The project owner confirmed that MAIA-2 was completed once before the experiment and spatial-frame ratings after each intervention; this timing is supplied by the owner rather than reconstructed from CSV timestamps. MAIA-2 assesses eight self-reported interoceptive dimensions (Mehling, Acree, Stewart, Silas, & Jones, 2018; Mehling et al., 2012); such reports should be distinguished from objective interoceptive accuracy and metacognitive awareness (Garfinkel, Seth, Barrett, Suzuki, & Critchley, 2015). Scores on the administered 0–5 response scale were independently recomputed using the final MAIA-2 item map: Noticing, items 1–4; Not-Distracting, 5–10; Not-Worrying, 11–15; Attention Regulation, 16–22; Emotional Awareness, 23–27; Self-Regulation, 28–31; Body Listening, 32–34; and Trusting, 35–37. Items 5–10, 11, 12 and 15 were reversed as $5 -$ response. Each subscale required all constituent items; missing responses were not imputed and no total score was formed. Recomputed and stored means were compared within 10^{-6}. The release does not supply the administered language record, repeated MAIA scores, or separate body-boundary and small-self fields.

This questionnaire extension is post hoc. Its scoring, estimands and test family were recorded before computing the new associations, but after inspecting the dataset and prior behavioural results. The original registration endpoint returned HTTP 403 during this review; its exact questionnaire wording could not be independently checked. The current repository Methods and analysis contract supply the available account of the plan, with the distinction between registered behavioural questions and this exploratory extension retained. The original registration and administered instrument records require verification before

submission.

We reported item frequencies, missingness, subscale descriptives, corrected item–rest correlations and raw Cronbach alpha. Sample-specific reliability can differ by dimension and occasion (Eggart, Todd, & Valdés-Stauber, 2021); alpha also has restrictive assumptions (McNeish, 2018). We therefore reported pointwise percentile 95% intervals from 2,000 whole-participant bootstrap draws for means and alpha, without treating a cutoff as evidence of validity, deleting items, or fitting a new 37-item factor structure in this small sample. No clinical change thresholds or pre/post tests were transferred from external validation studies.

Spatial-frame ratings are relevant to experiential changes studied in mindfulness research (Hanley et al., 2020), but the present release includes only condition-labelled A–H responses and corresponding codes 1–8. We verified agreement of these encodings and retained their ordinal order. Category distributions and the full paired transition table were reported. A two-sided exact binomial sign test compared higher versus lower Viscereality ratings among non-ties, with a Clopper–Pearson interval for the higher-rating proportion. Ties remained in the overall paired denominator and in paired dominance, defined as $(n_{\text{higher}} - n_{\text{lower}})/n_{\text{pairs}}$, with a participant-bootstrap interval. Order-stratified counts were descriptive. Equal spacing of categories was not assumed. Because the administered pictogram key is unavailable, higher codes were not labelled allocentric.

The retained supplementary association set comprised 33 Spearman correlations: eight MAIA-2 dimensions with each of Control rating, Viscereality rating, rating-change direction $(-1, 0, +1)$, and the equal-participant raw-RT Condition × Phase SOA-slope contrast; plus rating-change direction with that behavioural contrast. The slope contrast used the five cell means and primary SOA scaling; it does not estimate a sigmoid boundary. Complete pairs, average ranks, 19,999 pairing permutations, and the absolute-statistic two-sided tail with the plus-one correction supplied p values. All 33 association p values were reported without multiple-testing correction. This reporting choice replaced the initially specified Holm adjustment after inspecting the results; the analyses remain exploratory. The plus-one rule is part of Monte Carlo permutation inference, not adjustment across tests. Pointwise 95% intervals used 2,000 paired-person bootstrap draws with re-ranking, excluding and counting undefined draws. Legacy resampling used seed 20261116. These intervals are not simultaneous, and permutation inference assumes exchangeable participants under independence without covariate adjustment. The spatial condition comparison formed a separate, single exploratory test family. Inter-subscale correlations were descriptive without further testing. Missing questionnaires never excluded a person from the primary RT cohort; boundary-related correlations retained the original fit gate.

The main spatial follow-up was subsequently specified as the signed numerical difference $\Delta_i = Y_{i,\text{Viscereality}} - Y_{i,\text{Control}}$, regardless of presentation order. These are two post-intervention ratings, not change from a spatial baseline. This exploratory amendment was recorded before computing the new delta analyses but after inspecting the earlier questionnaire results. All eight continuous MAIA-2 dimensions were correlated separately with numerical Δ using the same permutation and paired-bootstrap procedure, with an independent generator seeded 20261117. One ancillary correlation linked Δ to the raw-RT SOA-slope interaction. The eight existing direction-only correlations were retained as ordinal sensitivities; the complete earlier set remained supplementary. Numerical subtraction assumes comparable category steps, and ranking the resulting differences does not eliminate that assumption.

Condition $\times$ pre-experiment MAIA moderation was estimated separately for each dimension by OLS regression of participant Δ on an intercept and mean-centred MAIA. A second specification added centred presentation order (0=Control-first, 1=Viscereality-first). Differencing eliminates participant intercepts; in this complete two-condition design, the MAIA slope tests the corresponding repeated-measures interaction, while the order coefficient represents Condition $\times$ Order. Independent long-format participant-intercept fits verified the equivalence. Slopes are code steps per one MAIA point. Ordinary two-sided t inference used $n - k$ residual degrees of freedom and interaction $F = t^2$ with $(1, n - k)$ degrees of freedom. This inference assumes a linear conditional mean, independent participants and homoskedastic Gaussian errors. HC3 covariance with t_{n-k} tails supplied approximate heteroskedasticity-sensitive intervals and p values. Both specifications and both uncertainty estimates were reported for every dimension, without multiple-testing correction. Maximum leverage and Cook distance were exported; no participant was removed on these diagnostics. Unknown order excludes a record only from the order sensitivity. Baseline MAIA can moderate the condition contrast; its single pre-experiment measurement cannot mediate an intervention-induced change in MAIA. No mediation analysis was fitted, and order adjustment does not establish absence of carryover.

2.8 Statistical analysis

The analysis plan was registered on AsPredicted (`https://aspredicted.org/az9f89.pdf`). The browser implementation follows the registered questions as operationalised in this Methods section, with a fixed executable contract in `workbook/ANALYSIS_PLAN.md`. It reads the released tables containing 42 participant records and treats participant ID as the repeated-measures unit. An audit subsequently identified two linked behavioural recordings with different questionnaire responses; full-release independence is therefore provisional

pending source mapping. Audio–Tactile and Baseline hits with recorded RT are pooled within person to calculate the mean and sample SD; values within mean ± 2.5 SD, inclusive, enter RT analyses. Participants with Catch false-alarm rate greater than 30% are excluded; low hit-rate flags are descriptive. A 200-ms lower cutoff is a sensitivity, not an additional primary rule. The partial session is retained primarily, with explicit exclusion sensitivities. Interactive preparation controls do not change this analysis population.

The primary question uses a Gaussian ML mixed model of individual Audio–Tactile RTs, with Condition, instructed Phase, numeric SOA, all interactions, and a participant random intercept. Control/Viscereality and Exhale/Inhale are coded $-.5/+.5$; SOA is centred and divided by its trial-level sample SD. A likelihood-ratio test compares the full model to the identical model without the three-way term. The secondary model tests Phase × SOA within Control. Five-level SOA coding is a sensitivity because coding was not fully specified in the registration. Random-intercept ML is profiled exactly using participant sufficient statistics and Woodbury identities, including the zero-variance boundary; it is independently checked against native `statsmodels` ML fits. Model-based normal Wald intervals condition on estimated covariance parameters and are not small-sample-corrected intervals.

Sensitivities include a participant random SOA slope; linear and log-RT regressions with participant-cluster-robust sandwich covariance; equal-participant contrasts of slopes through five SOA means; complete-person $2 \times 2 \times 5$ repeated-measures ANOVA with effect-specific Greenhouse–Geisser correction; condition-order subgroup descriptions; participant-cluster-robust binomial hit-probability regression; and RT regression with within-condition trial position and Condition × Phase-specific time trends. The random-slope covariance is estimated by Cholesky parameterisation with L-BFGS-B and Powell candidates and a zero-slope boundary. Cluster covariance uses the $G/(G-1)\,(n-1)/(n-p)$ correction and normal Wald inference. The participant-slope sign-flip test uses all sign patterns for at most 16 people and otherwise 19,999 Monte Carlo draws with seed 20261015. These analyses probe assumptions rather than replace the primary estimand.

The post hoc missing-RT stress tests assign each hit-labelled Audio–Tactile trial lacking an RT its observed participant × Condition × Phase × SOA cell median. Signed shifts of 0, ±100, and ±200 ms are then applied according to the three-way-contrast direction, and the RT trim and primary model are rerun. These scenarios do not impute misses or identify the missingness mechanism.

The boundary analysis requires trial-level sigmoid $R^2 \geq .70$ in all four participant Condition × Phase cells. The registration did not fully specify the four-parameter form, bounds, trial-versus-mean R^2, or whether this exclusion also applied to the raw-RT

interaction. We explicitly apply it to boundary inference and report the raw-RT interaction for the roster-based cohort. The transparent reimplementation is $f(x) = a + (b - a)\,\mathrm{expit}((x - m)/s)$, with $a, b \in [-2000, 2000]$ ms, midpoint m within the sampled distance range, and width $s \in [.1, 200]$ cm, using three starts. These are disclosed computational choices, not recovered historical optimizer settings. Registered approximate distances and trajectory-implied positions are fitted separately, after subtracting the participant Condition/Phase mean Baseline. Trial objectives use exact SOA sufficient statistics. We additionally calculate the saturated five-SOA trial-level R^2 ceiling, which bounds any deterministic SOA curve independently of sigmoid bounds. In-sample sigmoid/linear RMSE, five-mean fits, group fits, and parameter-bound hits are descriptive. Four-parameter fits to five means have one residual degree of freedom and cannot rescue failed trial gates. Boundary midpoint/sharpness and related MAIA-2 or pictographic associations require four qualifying, bound-free trial fits per person; otherwise these outcomes are explicitly not estimable.

A separate post hoc extension examined whether one common procedure could describe four group RT profiles with interpretable sigmoid transitions. Audiotactile and related visuotactile work motivates examining distance-dependent transitions (Canzoneri et al., 2012; Noel et al., 2020), while expectancy-controlled logarithmic patterns and critical experimental evidence make the response form an empirical question (Hobeika, Taffou, Carpentier, Warusfel, & Viaud-Delmon, 2020; Holmes et al., 2020). This extension addresses group-profile description; it cannot recover participant boundaries that fail the inference gate. Candidate families were developed after inspecting the data and were not registered. Reporting the full finite grid follows the transparency rationale of multiverse analysis (Steegen, Tuerlinckx, Gelman, & Vanpaemel, 2016), without claiming an exhaustive set of reasonable analyses or treating the most sigmoid-looking result as a confirmatory finding.

Kernel local-constant and local-linear regression provide established signal-smoothing estimators (Li & Racine, 2007), while local weighted regression, curvature regularization, and isotonic projection provide alternative smooth or shape-constrained approximations (Busing, 2022; Cleveland, 1979; Wahba, 1990). The four-parameter logistic supplies descriptive location and scale parameters (Ritz, Jensen, Gerhard, & Streibig, 2019); whether the measurements identify them is a separate question from whether the curve fits well (Raue et al., 2009). These sources motivate the estimation tools, not particular bandwidths, bounds, or a requirement that PPS observations follow a sigmoid. The companion's miniature literature review maps each choice to supporting and contrary evidence and records the limits of source access.

The extension used the same retained Audio–Tactile trials and five design positions.

Within each participant × Condition × Phase × position cell, five extraction methods were considered: arithmetic mean, median, 10% and 20% per-tail trimmed means, and 20% per-tail winsorized means. Winsorization replaces the lowest and highest $\lfloor .2n \rfloor$ values by the adjacent retained order statistics; it does not remove further trials. The resulting cells were averaged with equal participant weights. Robust summaries probe the influence of extreme RTs and cutoff choices (Ratcliff, 1993); the specific percentages remain exploratory. Mean, median and winsorized summaries target different summaries of the response distribution, and the midpoint of a curve through group summaries is not generally the mean of individual midpoints. Thirty-two smoothing specifications crossed Gaussian local-constant and local-linear bandwidths, tricube local-linear bandwidths, discrete curvature penalties, isotonic regression, and Gaussian smoothing followed by isotonic projection, including an unsmoothed comparator. Separate versus common widths across the four curves gave $5 \times 32 \times 2 = 320$ pipelines. Tricube smoothing here is a fixed-bandwidth local-linear implementation without robust residual reweighting; the discrete penalty is not an exact cubic smoothing spline. The complete grid is executable in `descriptive_sigmoid.py`.

Distances were normalized to $[0, 1]$ using the registered 10–100 cm range. Each candidate fitted $a + A\,\mathrm{expit}((d - m)/s)$ with $A \in [0, 500]$ ms, $m \in [10, 100]$ cm and $s \in [4.5, 90]$ cm. Increasing RT with distance was imposed as a descriptive shape constraint, allowing $A = 0$ rather than forcing a nonzero transition. Offset and amplitude were profiled analytically over 61 equally spaced midpoints and 35 geometrically spaced widths. The selected method used a finer 241-by-81 grid. The width floor prevents transitions substantially narrower than the smallest sampled spacing: its 10–90% span is 19.8 cm, versus a minimum 17-cm gap. These bounds are explicit regularization choices and were expanded in a sensitivity analysis. No display interpolation was treated as an additional observation.

The same fixed six participant folds (seed 20261016) and five omitted positions evaluated every candidate. For each participant fold and position, extraction, group averaging, smoothing and fitting used only the other participants at the other four positions. The predicted held-out value was centered on the training curve's four unsmoothed extracted values; the fixed scoring target was each test participant's raw cell mean at the omitted position minus that participant's raw mean across the other four positions. Squared errors were averaged over participants, groups and held-out positions. This common shape target permits comparison of mean and robust extraction despite their different overall RT levels. The pipeline with minimum RMSE was selected uniformly for all four curves. Constant and linear benchmarks used identical folds and targets. These reused tuning scores do not constitute an independent post-selection test. Evaluation conditions on profiles cleaned beforehand: participant-wide trimming thresholds use all eligible AT and Baseline hit RTs,

including the position later omitted from fitting. Thresholds are not shared across people, but this is not end-to-end prospective prediction with preprocessing learned without the omitted position. The target measures relative profile shape, not absolute RT accuracy.

Uncertainty was described with 1,000 participant-cluster bootstrap draws (seed 20261017), retaining all four curves and all five positions together within each sampled person and refitting the selected smoother and logistic model (Efron & Tibshirani, 1993). Within-person extraction is deterministic and can be cached before resampling. Pointwise curve bands include flat fits. Midpoint, width and slope percentile intervals condition on nonflat fits and on the selected method; flat fractions and parameter-bound fractions are reported. Method selection is not repeated within this bootstrap. Paired parameter contrasts use the same resamples and receive no confirmatory p values. Fixed trimmed-mean/kernel alternatives, trajectory-implied coordinates, expanded bounds, and the full candidate table expose method dependence. Subtracting a participant/Condition/Phase constant Baseline would only shift these translation-equivariant curves vertically; no same-label cue-time matching was assumed.

An additional exploratory shape audit required at least the fitted 10–90% transition within the measured range, two observed positions strictly on each side of the midpoint, one observed position within the 20–80% transition, and maximum deviation from a best-fitting line on a 181-point display grid of at least 5% of the model's observed-range change. Positive amplitude was required to classify a transition as visible; flat fits remained allowed. These are explicit post hoc display and sampling criteria, not a validated definition of PPS. The same 320 pipelines were rerun on the restricted grid, and all 160 extraction/smoother combinations were also fitted with a line. Constraints use known design coordinates only, including when a position is omitted; no held-out outcome enters smoothing or fitting. Scores are weighted by the number of test participants, with the same raw-mean contrast target for every family.

To reduce the optimism of reporting a tuning minimum as evaluation evidence (Varma & Simon, 2006), six outer participant folds (seed 20261026) each repeated full selection among the candidates of each family using five inner participant folds crossed with five omitted positions. The selected pipeline then predicted outer test participants with each target position omitted from fitting. Inner selection uses all five positions from training participants; this evaluates new people at known design locations, rather than generalization to an unseen experimental design. A zero-contrast constant provides a fourth benchmark. Families and thresholds were developed post hoc, and training sets overlap; no significance test treats folds as independent replicates. Nested evaluation does not remove all sampling variability (Varoquaux et al., 2017); small RMSE differences are interpreted descriptively. The restricted

family's final estimates also receive 1,000 whole-participant bootstrap refits, conditional on its selected method and constraints. For display, each draw and its corresponding extracted data are centered on the mean across the five sampled positions. Centering is an additive shift and preserves curvature. A separate geometry-only figure normalizes fitted amplitudes, with no observations or uncertainty represented on that scale.

As an independent exploratory check on the observations, raw unsmoothed participant means were projected onto three orthonormal contrasts orthogonal to an intercept and distance. One-sample Hotelling T^2 tests assess any nonlinear component, under independent participants, multivariate-normal contrasts and nonsingular covariance. The $F(3, n-3)$ transformation was independently checked against `statsmodels`; Holm correction covers the four curves. This is not a sigmoid-specific test. The PPS literature motivates investigating distance-dependent transitions (Canzoneri et al., 2012; Noel et al., 2020), but also reports expectancy-sensitive logarithmic patterns (Hobeika et al., 2020) and methodological concerns about flexible analysis and small effects (Holmes et al., 2020). The focused literature review used indexed abstracts for the PPS and validation papers and official software or author sources for implementation references. Full-paper retrieval and current-literature coverage were limited; the companion records exact searches, access levels and source hashes. No source is taken as evidence that this dataset must be sigmoid.

Supplementary paired contrasts cover Baseline Condition, Phase and their interaction, overall Audio–Tactile-minus-Baseline RT, far-minus-near patterns, same-label Baseline comparisons, and pink-minus-white RT. Their ten unadjusted p values are accompanied by Holm-adjusted values for one family. A four-noise Friedman comparison includes only people with all four sounds available, without assigning acquisition-version labels. Paired contrasts have 95% t intervals; null detections are not equivalence tests. Same-label Baseline subtraction is descriptive because nominal labels do not establish matched tactile-cue times.

Exploratory individual sigmoid proxies and common plateaus. After the original fit gate failed, an owner-requested extension sought four Condition × Phase descriptive curves per participant record. The model $b + a\,\mathrm{expit}((x-m)/w)$ allowed signed amplitude $a \in [-500, 500]$ ms and zero amplitude; decreasing responses were not reversed. Distances were scaled to $[0, 1]$. The same measured-range visibility constraints defined the primary shape grid; a broad grid, straight lines and constants supplied benchmarks. Three cell summaries (mean, median, 20% per-tail trimmed mean), four smoothing choices (none; Gaussian local constant at .15 or .30 span; Gaussian local linear at .30) and penalties 0, 1 or 4 yielded 36 visible, 36 broad and 12 linear methods. All participants used one selected

visible method, chosen on RT prediction without questionnaire outcomes. The finite grid contained 61 midpoint and 35 width values before visibility restriction; a 241-by-81 grid supplied a resolution sensitivity.

For each cell, the objective divided squared residual error by the mean working variance of its cell means (floor 1 ms^2) and added $\lambda[((m - m_0)/.25)^2 + (\log(w/w_0)/\log 2)^2]$. The prior centre was the unpenalized same-cell pooled shape excluding all copies of the target recording. This is a penalized point estimator, not Bayesian posterior inference. Hierarchical cognitive-model estimation motivates examining pooling and parameter recovery (Ahn, Krawitz, Kim, Busemeyer, & Brown, 2011); it does not validate these particular penalties for PPS. Offset and signed amplitude were profiled exactly on a common shape grid. Six outer participant/linked-record-group folds repeated method tuning by omitted-position prediction; outer test fits used the other four positions and a pool derived only from training records. The common scoring target was each omitted raw cell mean minus the mean over the other four positions. Earlier participant-wide cleaning remained fixed. Linked recordings were kept together in folds and excluded together from each other's pooling prior.

A post-results sensitivity shared b and a across each person's four curves while retaining separate midpoint and width. Alternating grid-shape and weighted linear updates used three signed starts, a 60-iteration limit and recorded convergence; this is not a guarantee of the global optimum. The same selected visible preprocessing and penalty were used. Shared plateaus reduce nuisance parameters but impose common amplitude and direction. For either model, normalization was $(RT - \min(b, b + a))/|a|$: lower and upper fitted plateaus are 0 and 1, midpoint and width are unchanged, and observed values are not clipped. Zero amplitude makes normalization undefined; amplitude intervals including zero flag unstable scales. Normalized steepness magnitude is $1/(360w)$ per cm and signed RT slope is $a/(360w)$ ms/cm; midpoint is $10+90m$ cm. Normalized panels show model geometry, with raw observations and their standard errors separately displayed.

Five hundred within-cell trial resamples refitted both models for conditional parameter intervals. A separate 500 record-group-plus-trial bootstrap refitted pooling and parameters for correlation intervals; linked records stayed in the same sampling group, although their ambiguous trial mappings cannot be repaired by this procedure. Selection, original cleaning and constraints remained fixed. Serial dependence within cells and method-selection uncertainty are not covered. Ten random stratified split halves assessed within-session rank stability, without claiming test–retest reliability (Hedge, Powell, & Sumner, 2018). Thirty simulations per scenario added resampled empirical centred trial residuals to flat, linear, known 80-ms sigmoid, or common-plateau sigmoid signals. Recovery and error were reported

for both models; interval coverage was not estimated.

The main 27 links per model correlated each of eight baseline MAIA-2 dimensions with each of three Viscereality–Control parameter differences (equal phase weights), plus spatial-code difference with each parameter difference. Ninety-six cell-specific MAIA correlations and three spatial-direction sensitivities brought the total to 126 per model. All used Spearman coefficients and 19,999 pairing permutations; p values condition on point proxies while refit-bootstrap intervals propagate additional parameter uncertainty. No multiplicity correction was applied. The P011/P033 linkage discovery prompted three explicitly exploratory exclusions: both records, only P011, and only P033. Each repeated method selection and both sets of 126 associations, with point estimates, unadjusted p values and 500 additional hierarchical refit-bootstrap intervals per exclusion, yielding 1,008 tests overall. The full-release RT–questionnaire analyses remain provisional; none of these exclusions establishes the correct recording identity. They do not replace the original gated boundary hypotheses.

Remaining-analysis robustness amendment. After inspecting the earlier results, a finite amendment recomputed cleaning and primary/Control-only numeric and categorical ML comparisons, raw/log cluster regressions, equal-person slopes and raw-RT repeated-measures ANOVA for each recording exclusion. Equal-phase Condition, equal-condition Phase and Condition $\times$ Phase contrasts were calculated for all three individual parameters under both plateau models. Each cohort's 500 hierarchical refits supplied pointwise intervals on mean contrasts and questionnaire links; all three contrasts were also evaluated in the existing split halves and known-signal simulations. Derived contrasts and fitted scores used in associations were rounded to 12 decimal places before ranking, preventing machine-precision cancellation from splitting mathematical ties; point-parameter tables, means and prediction errors retained original precision. Method selection, original cleaning and constraints remained fixed within these individual-parameter resamples.

The cohort excluding both ambiguous recordings additionally compared a log-distance response $a + b\log(d)$, with 160 extraction/smoothing options matching the linear family, against the other four group-response families. Smoothing retained physical-distance coordinates. All families used the same six-outer/five-inner participant/omitted-position prediction target. This shape comparison does not reproduce an expectancy-controlled experiment. Nine uniform visibility specifications crossed width floors of 2.25, 4.5 and 9 cm with permissive, existing and strict criteria. These required respective tail fractions

.20/.10/.05, departures from a line of .025/.05/.10, one/two/two measured positions per flank, and at least one point within the 10–90/20–80/30–70% transition. The exclusion cohort's selected visible preprocessing remained fixed; all nine fits and descriptive leave-position errors were reported. Thirty whole-person resamples repeated the 320-method visible selection with repeated copies kept in a common validation fold (seed 20261019). These diagnose selection stability; they are insufficient for tail confidence intervals. All newly added tests are unadjusted. The amendment is post hoc and does not estimate serial-dependence effects, prospective preprocessing performance or physiological validity.

The registration named a Cauchy width of 0.707 but did not fully specify mixed-model priors and comparisons. No BIC approximation is presented as a registered mixed-model Bayes factor. Exploratory one-sample/paired Jeffreys–Zellner–Siow factors use a Cauchy $\sqrt{2}/2$ prior on standardized participant contrasts under a normal likelihood (Rouder, Speckman, Sun, Morey, & Iverson, 2009); Shapiro–Wilk diagnostics accompany them. All data-supported computations run in the browser using NumPy/SciPy in Pyodide, with versions, model status, retained data, and source/input/output hashes in the downloadable record. Source-dependent waveform, respiratory-compliance, date, version, and onset-definition checks are marked unavailable when the public CSVs cannot support them.

2.9 Respiratory compliance and participants

Respiratory phase was assigned in the trial schedule under the instructed 4–4–4–4 box-breathing pattern; trials were not gated online by the belt signal. The archived decoder identifies which prerecorded breathing instruction played and can reconstruct final phase labels from trial-order parity. These procedures identify the instructed half-cycle, not the participant's measured inhalation or exhalation. A respiratory belt was part of the acquisition protocol, but the current publication-pool trial table contains protocol phase labels rather than independently belt-verified phases. The available source recordings do not yet support a uniform trial-by-trial respiratory-compliance audit across all four acquisition versions. Phase-related effects must therefore be interpreted as effects of instructed phase under the compliance assumption.

The released roster contains 42 participant records for adults aged 19–37 years, with 41 complete session records and one partial session. Two records have identical behavioural observations with intervention labels reversed; their correct recording-to-person mapping requires source review. Each participant was scheduled for both VR conditions, and 21 completed each condition order. The registration disclosed eight already-collected

pilot participants and a stopping target of 30 complete datasets, 15 in each order. The released roster exceeds that target; dates and the reason for continued collection cannot be reconstructed from these public CSVs and require author documentation before submission. A balanced pooled order split does not establish balance within historical acquisition versions. One participant has behavioral PPS measures without the questionnaire battery. The project owner confirmed ethical approval and authorised public release of the deidentified data after removal of names and verification of a separate backup; approval-body details should be supplied by the authors for the final ethics statement.

3 Results

3.1 Released cohort and response quality

The analysis used all 42 participant records in the released tables (34 women, 8 men; mean age 23.0 years, SD 4.0, range 19–37). Twenty-one completed each condition order. There were 17,114 observed trials of 17,136 planned slots: 10,068 Audio–Tactile, 5,033 Baseline, and 2,013 Catch trials. Forty-one sessions were complete; one partial session lacked 22 trials but supplied all 20 Condition $\times$ Phase $\times$ SOA cells. There were 24 Catch false alarms, and no participant exceeded the 30% Catch exclusion rule. Low-hit-rate flags are reported without adding an exclusion criterion.

Among hit-labelled trials, 105 RTs were missing (80 Audio–Tactile and 25 Baseline). These trials remained in hit-rate denominators. The participant-wide mean $\pm$ 2.5-SD trim removed 382 of 13,694 recorded hit RTs, retaining 8,916 Audio–Tactile and 4,396 Baseline RTs. Adding a 200-ms minimum after trimming removed no further observations. Forty-one participants had the full questionnaire measures and one had PPS measures only.

A recording-integrity audit found that P011 and P033 share all 408 behavioural trial records, with Control/Viscereality labels swapped, despite different questionnaire responses. The CSVs were preserved; the correct recording-to-person mapping is unresolved. Full-release behavioural inference below is provisional and cannot assume 42 independent recordings. The new individual-parameter extension reports explicit exclusions of both records and of each separately. Questionnaire-only spatial/MAIA results do not depend on the RT mapping.

3.2 Primary interaction and sensitivity analyses

Adding the Condition × Phase × numeric-SOA term to the trial-level random-intercept ML model did not improve fit, $\chi^2(1) = 0.024$, $p = .878$. With symmetric Condition/Phase coding and standardized SOA, its estimate was 0.81 ms per SOA SD (SE 5.31; 95% model-based Wald CI $[-9.60, 11.23]$). This interval permits effects in either direction and does not establish equivalence.

The categorical-SOA comparison gave $\chi^2(4) = 2.27$, $p = .686$. Excluding the partial session left 41 participants (numeric $p = .807$; categorical $p = .773$). The Control-only Phase × SOA test was also not detected (numeric $p = .430$; categorical $p = .468$). Allowing a participant random SOA slope yielded $p = .859$. The participant-cell repeated-measures three-way interaction was $F(4, 164) = 0.99$, Greenhouse–Geisser $\epsilon = 0.839$, corrected $p = .406$.

The equal-participant five-SOA slope contrast was 0.03 ms per SOA SD, 95% CI $[-7.88, 7.95]$, $t(41) = 0.008$, $p = .993$. The Monte Carlo sign-flip test used 19,999 draws with a fixed seed ($p = .994$). Its exploratory paired JZS factor was $\mathrm{BF}_{10} = 0.167$ ($\mathrm{BF}_{01} = 6.00$). This normal-likelihood participant model is distinct from the registered mixed model; the slope distribution departed from normality (Shapiro–Wilk $p = .005$).

The exploratory participant-cluster-robust models gave the following three-way terms. Normal Wald intervals use the documented finite-cluster sandwich correction.

Model	Estimate	95% CI	p
Raw RT	-1.543	$[-9.57, 6.49]$	.706
Log RT	-0.003	$[-0.01, 0.01]$	.614
Time-on-task-adjusted RT	-1.584	$[-9.59, 6.42]$	.698
Hit probability (log odds)	0.001	$[-0.23, 0.24]$	.994

The missing-hit-RT stress tests imputed only the 80 Audio–Tactile hits lacking RTs, then recomputed trimming. Median-only imputation gave $p = .960$; signed -200 and $+200$ ms scenarios gave $p = .428$ and $p = .253$, respectively. These specified scenarios do not recover missed-trial RTs or identify the missingness mechanism. Full model coefficients, scaling, diagnostics, intermediate tables, and all stress scenarios are in the companion's downloadable report.

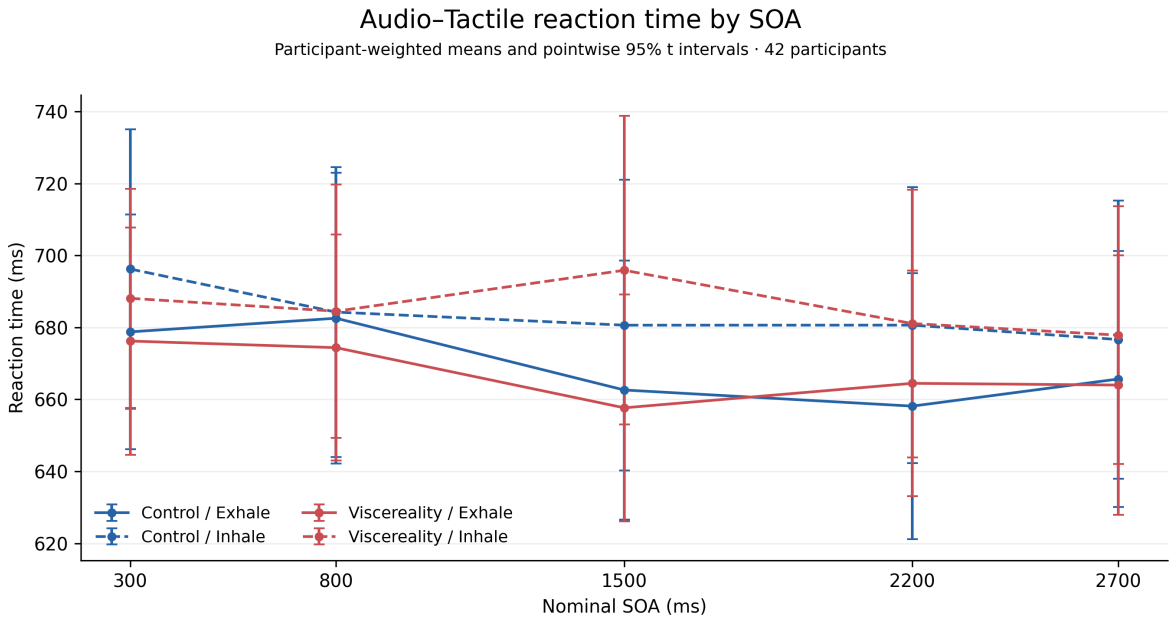


Fig. 3 Audio–Tactile RT by SOA, Condition, and instructed respiratory phase in the 42-person cohort. Each point averages participant means with equal person weight; all cells contain 42 participants. Intervals are pointwise 95% t intervals across participant means. Longer SOAs correspond to nearer nominal auditory positions. Lines connect means and are not sigmoid fits; inference comes from the stated models.

3.3 Sigmoid viability and descriptive PPS shape

None of 168 trial-level participant × Condition × Phase sigmoid fits reached $R^2 \geq .70$ (median $R^2 = 0.038$; maximum 0.419). The largest saturated five-SOA trial-level R^2 ceiling was 0.430. Thus no deterministic sigmoid of these SOAs could reach .70 in any cell, independent of the particular nonlinear optimizer or parameter bounds. Sigmoids improved unpenalized in-sample RMSE over a line in 164/168 cells; this does not establish stable boundaries.

Fitting four parameters to five SOA means instead yielded 79/168 qualifying cells and 3 participants passing all four cells, with none having four bound-free fits. Such fits have one residual degree of freedom and were not used to rescue the failed trial-level gate. Replacing approximate registered distances with trajectory-implied positions left the trial-level pass count at zero. Parameter-bound diagnostics and group fits are supplied descriptively. Participant midpoint/sharpness contrasts and their MAIA-2 or pictographic associations were not estimable under the planned fit gate.

3.4 Four uniformly estimated descriptive sigmoid curves

A separate post hoc analysis obtained nonflat sigmoid approximations for all four Condition × Phase groups using a single extraction, smoothing, and fitting rule (Figure 4). All 42 participants contributed all five positions with equal person weight. The search compared 320 pipelines spanning mean, median, trimmed-mean and winsorized summaries; Gaussian local-constant/local-linear and tricube local-linear smoothing; curvature penalization; isotonic projection; and separate versus shared logistic widths. The original trial-level gate above was retained as a distinct analysis.

The lowest crossed participant/position prediction score selected 20% per-tail winsorization within participant cells, Gaussian local-constant smoothing with bandwidth 0.5 of the nominal distance span (45 cm), and separate logistic widths. This same pipeline was applied to all four curves. The tuning RMSE was 47.053 ms, compared with 47.317 ms for a line and 47.437 ms for a constant. The fixed prediction target was each held-out person's raw mean at the omitted position minus that person's raw mean across the other four positions. The small advantage over a line is a post-selection descriptive score, not independent evidence of sigmoid superiority.

For $f(d) = a + A\,\mathrm{expit}((d-m)/s)$, m is the half-amplitude midpoint, s is the logistic width, and the RT derivative at the midpoint is $A/(4s)$. Smaller widths indicate sharper normalized transitions; RT slopes additionally depend on amplitude. Width is not a Gaussian standard deviation. Estimates refer to the registration's approximate distance labels.

Condition / phase	m (cm)	s (cm)	$f'(m)$ (ms/cm)
Control / Exhale	71.50	35.29	0.0907
Control / Inhale	96.25	42.56	0.0846
Viscereality / Exhale	82.75	35.29	0.0900
Viscereality / Inhale	25.38	30.38	0.0686

The 1,000 whole-participant bootstrap resamples preserved each person's four curves together and refitted smoothing and the sigmoid model. Conditional 95% percentile intervals for the four midpoints, in table order, were [37.6, 88.0]; [18.5, 100.0]; [31.8, 100.0]; [10.0, 93.3] cm. Flat fits occurred in 0.9%, 8.7%, 3.2%, 8.7% of resamples; midpoint-bound hits, including flat draws, occurred in 2.0%, 21.0%, 6.3%, 14.1%. Midpoint and width intervals exclude flat draws, for which these parameters are unidentified, and condition on the selected pipeline. They do not include uncertainty from choosing among 320 methods and should not be treated as unconditional coverage guarantees.

Descriptively, the Inhale-minus-Exhale midpoint difference was 24.75 cm in Control and

-57.38 cm in Viscereality. Their difference was -82.12 cm, conditional bootstrap interval [-107.25, 25.62] cm, based on 828 resamples with all contributing curves nonflat. No confirmatory test was assigned to this post hoc group-parameter contrast. The report includes all paired midpoint, width and RT-slope comparisons.

The fitted 10–90% transition spans were 155.1, 187.0, 155.1, 133.5 cm, each exceeding the sampled 90-cm span. Consequently, the complete S shapes and their asymptotes depend on extrapolation. A fixed alternative using 20% trimmed means and a narrower Gaussian bandwidth of 0.3 yielded midpoints 61.7, 84.6, 68.1, 31.4 cm, with tuning RMSE 48.784 ms. Applying the selected rule to trajectory-implied coordinates yielded 74.7, 87.3, 82.0, 34.3 cm; expanding the midpoint/width/amplitude search bounds yielded 70.8, 94.8, 84.2, 25.8 cm. These method and coordinate sensitivities limit a literal boundary interpretation. All candidate scores and parameter estimates, including edge and flat solutions, remain available in the companion.

3.5 Measured-range PPS shape and predictive support

The stronger exploratory goal required a visible transition within the observed range, rather than a sigmoid that is nearly linear over that range. None of the 320 original pipelines met all measured-range criteria in all four curves. An additional 320 pipelines imposed those criteria, while 160 linear pipelines received the same extraction and smoothing options. The lowest-scoring constrained method used 10% per-tail trimming, a Gaussian bandwidth of 13.5 cm, increasing isotonic projection and separate logistic widths. Its tuning RMSE was 47.086 ms, versus 47.178 ms for the tuned linear family. These are method-selection scores.

Condition	Phase	Midpoint (cm)	Width (cm)	Slope (ms/cm)	Width floor
Control	Exhale	56.12	4.50	0.9194	Yes
Control	Inhale	72.62	8.51	0.4234	No
Viscereality	Exhale	69.25	4.50	0.7733	Yes
Viscereality	Inhale	30.62	4.50	0.6310	Yes

Table 1: Imposed-shape descriptive estimates. A visible transition is enforced by the model constraints; it is not an empirical finding. Conditional intervals and constraint-hit frequencies are provided in the companion.

Three of the four width estimates reached the 4.5-cm minimum, and the Control/Inhale midpoint reached the upper sampling constraint. Conditional 95% midpoint intervals, in table order, were [44.50, 72.62]; [27.25, 72.62]; [38.11, 72.62]; [27.25, 72.62] cm. These intervals are truncated by the shape restrictions and exclude flat resamples; they do not establish stable physiological boundaries.

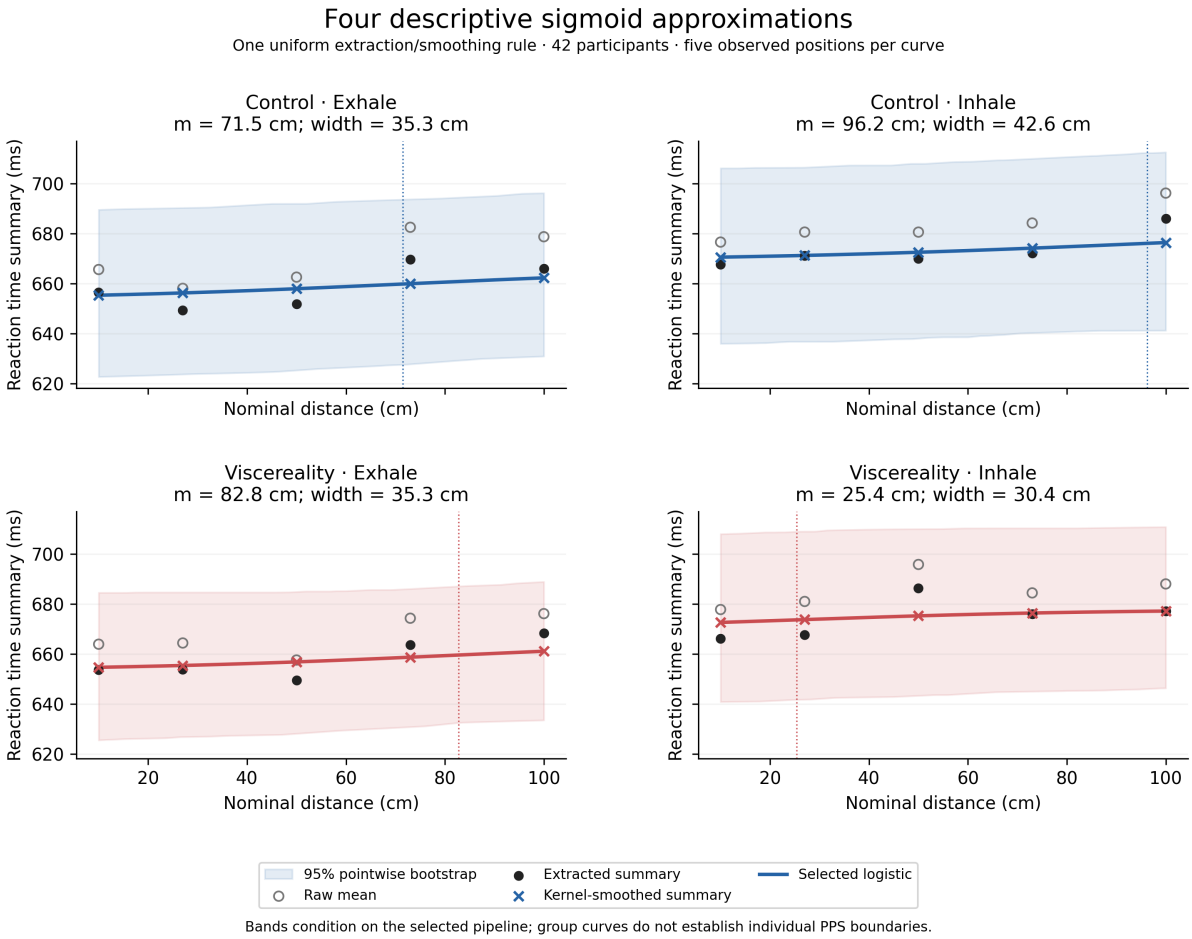


Fig. 4 Four uniformly estimated descriptive sigmoids. Open gray circles are raw participant-weighted means; black points are participant-weighted winsorized cell summaries; crosses show Gaussian-kernel-smoothed summaries. Solid curves are logistic approximations fitted to the five smoothed design points, with no interpolated observations added to the fit. Vertical dotted lines mark the fitted midpoints. Bands are pointwise participant-bootstrap percentiles conditional on the selected pipeline, including flat resamples. The full search, parameter uncertainty, and model extrapolations are available in the browser companion.

Six outer participant folds repeated the complete pipeline selection inside each training set, using five inner participant folds crossed with omitted positions. Outer RMSE was 47.772 ms for the original sigmoid family, 47.634 ms for the constrained family, 47.701 ms for tuned lines and 47.437 ms for a constant. The constrained curve's small advantage over tuned lines did not extend to the constant benchmark. These exploratory estimates address new participants at the known design positions; overlapping training sets and post hoc family development preclude a confirmatory model-superiority claim.

A separate unsmoothed participant-level test projected each five-position curve onto the three contrasts orthogonal to an intercept and distance. Hotelling tests of these nonlinear components gave Holm-adjusted p values of 0.215, 1.000, 1.000, 1.000, in table order. This test concerns any nonlinearity, not specifically a sigmoid, and assumes multivariate-normal participant contrasts. Failure to detect curvature does not prove linearity. Together, the

predictive results and active constraints leave the data-supported PPS-shape goal unresolved, despite obtaining four visibly S-shaped model approximations.

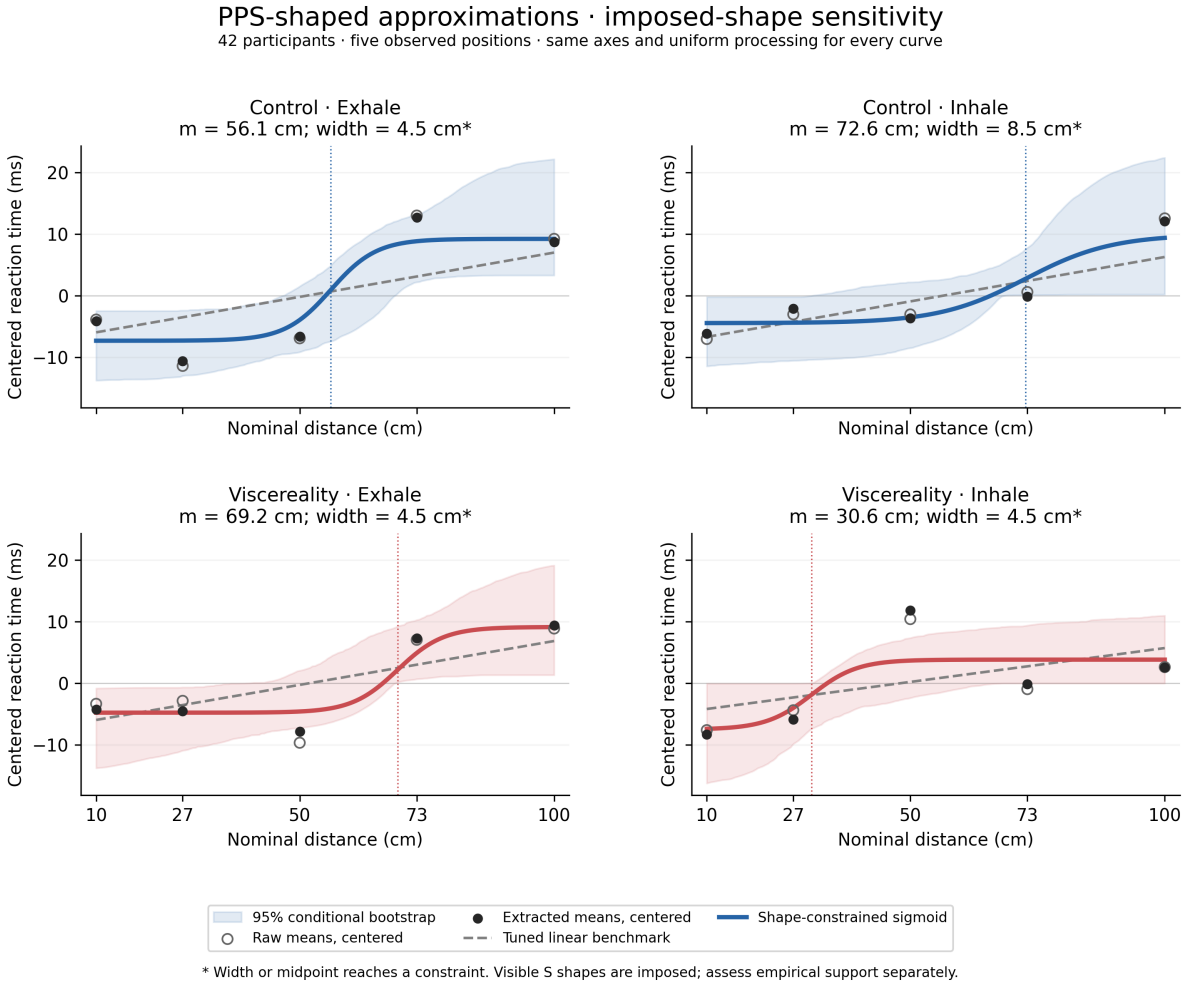


Fig. 5 Measured-range PPS-shaped sensitivity. Common linear axes preserve between-curve scale comparisons. Raw means (open circles), trimmed summaries (black points), fitted curves and the tuned line are each centered on their corresponding five-position mean; subtracting a constant cannot create curvature. Bands resample participants jointly, refit the constrained model and recenter each draw. They condition on the selected extraction and shape restrictions. Asterisks mark active width or midpoint constraints. All five observed positions remain visible.

3.6 Baseline, sound, and order diagnostics

Across ten supplementary paired contrasts, Holm adjustment was applied as one family. Overall Audio–Tactile minus within-Condition/Phase mean Baseline RT was -18.81 ms, 95% CI $[-27.00, -10.63]$, unadjusted $p < .001$, Holm $p < .001$. Baseline Inhale minus Exhale RT was 18.08 ms, 95% CI $[3.28, 32.88]$, unadjusted $p = .018$, Holm $p = .143$. The Baseline Condition × Phase contrast was 17.43 ms, 95% CI $[1.56, 33.30]$, unadjusted $p = .032$, Holm $p = .225$. These checks neither establish phase invariance nor provide adjusted evidence of

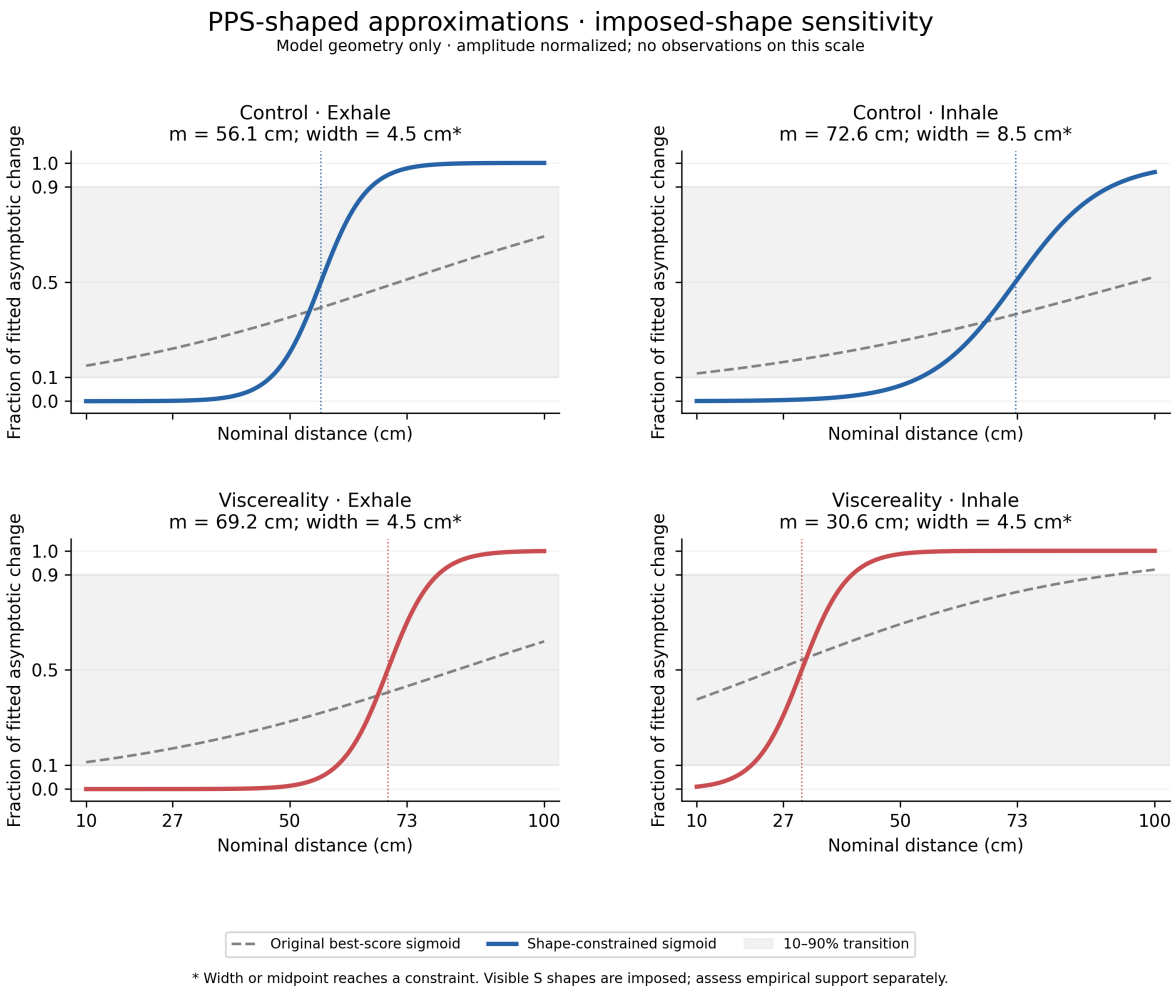


Fig. 6 Fitted shape separated from empirical support. The vertical axis normalizes each fitted asymptotic RT change to 0–1. Colored lines show the imposed-shape family; dashed gray lines retain the original best-score family. This diagnostic displays model geometry only, without observations or empirical uncertainty on this scale. It shows why the original curves did not display both plateaus within the measured range, and how explicit constraints produce an S shape. It is not additional evidence for that shape in the measurements.

a Baseline phase difference.

The far-minus-near Audio–Tactile difference was 13.80 ms, 95% CI $[-1.82, 29.42]$; the analogous Baseline nominal-label difference was 11.69 ms, 95% CI $[-6.90, 30.28]$. Their difference was 2.11 ms, 95% CI $[-18.44, 22.66]$. Far Audio–Tactile minus Baseline was -10.50 ms using the Condition/Phase reference and -24.83 ms using the same nominal SOA label (Holm $p = .023$ for the latter). Equal labels do not establish cue-time matching; these differences cannot isolate auditory proximity or remove temporal expectancy.

Pink-minus-white Audio–Tactile RT was -2.86 ms, 95% CI $[-9.30, 3.58]$. Among the 20 participants with all four noise types available, the exploratory Friedman test gave $\chi^2(3) = 7.80$, $p = .050$. This availability-defined subset cannot be assigned to an acquisition version from the released tables. The two order subgroups each contained 21 participants;

their slope contrasts were -4.19 ms, 95% CI $[-14.64, 6.27]$, and 4.25 ms, 95% CI $[-8.27, 16.78]$. These descriptive subgroup results do not establish order equivalence.

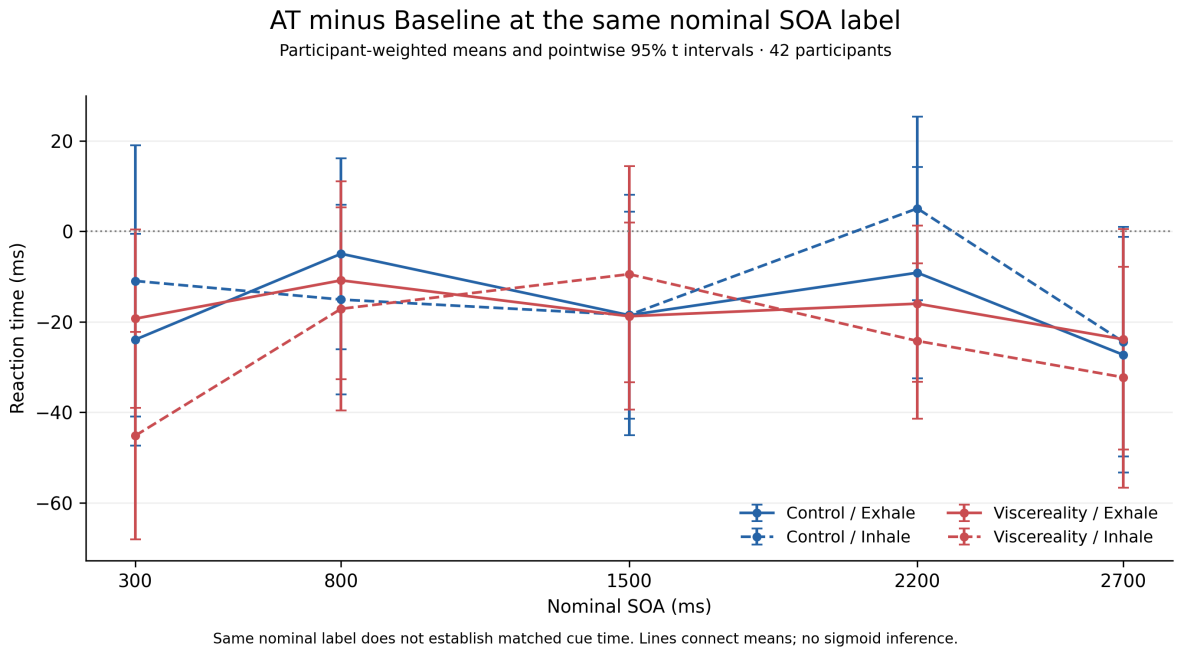


Fig. 7 Exploratory Audio–Tactile minus Baseline RT paired by nominal SOA label. Points average 42 participant differences within Condition × Phase × label. Negative values indicate faster Audio–Tactile responses. Baseline has no sound; matching labels does not establish equal trial-relative tactile-cue time. Intervals are pointwise 95% t intervals across participants. Lines connect means and carry no boundary inference.

3.7 MAIA-2 and spatial frame of reference

This post hoc extension analysed 41 complete MAIA-2 records and 41 complete spatial-frame pairs. All 328 stored MAIA-2 subscale scores matched independent recomputation from the 37 items, including reverse scoring; there were 0 mismatches. The participant with missing questionnaires remained in the behavioural cohort. Table 2 reports all eight dimensions and internal-consistency uncertainty. These are self-reported interoceptive dimensions, not objective interoceptive accuracy. The lower alpha estimates for Noticing and Emotional Awareness warrant caution; no items were removed to improve reliability.

Viscereality spatial-frame ratings were higher than Control for 22 participants, lower for 6, and tied for 13 (Fig. 9). The two-sided exact sign test gave $p = 0.00372$. Among the 28 non-ties, the higher-rating proportion was 0.786, exact 95% CI $[0.590, 0.917]$. Paired dominance across all 41 pairs was 0.390, percentile 95% CI $[0.171, 0.610]$. Both condition medians were category D. Higher/equal/lower counts by condition order were Control-Viscereality: 13/6/2; Viscereality-Control: 9/7/4. These descriptive order counts do not rule out period or carryover effects. The administered pictogram key is absent from the

Dimension	Mean	SD	Alpha	95% alpha interval
Noticing	3.24	0.66	0.565	[0.188, 0.740]
Not Distracting	2.24	0.85	0.842	[0.725, 0.906]
Not Worrying	2.29	0.82	0.704	[0.511, 0.808]
Attention Regulation	2.96	0.67	0.784	[0.648, 0.858]
Emotional Awareness	3.47	0.70	0.589	[0.314, 0.740]
Self Regulation	2.85	0.86	0.814	[0.690, 0.885]
Body Listening	2.80	0.92	0.777	[0.602, 0.867]
Trusting	3.59	0.98	0.861	[0.777, 0.909]

Table 2: MAIA-2 subscale scores (0–5), each with $n = 41$. Intervals are pointwise percentile intervals from 2,000 participant-bootstrap draws. Alpha is a sample internal-consistency diagnostic, not a test of dimensionality or validity.

release, so a higher A–H code is not assigned an allocentric or egocentric interpretation. This exploratory ordinal difference does not establish a PPS-boundary change or its mechanism.

MAIA-2 was administered once before the experiment and spatial-frame ratings after each intervention, as confirmed by the project owner. The main follow-up outcome was the signed numerical code difference Δ = Viscereality − Control, irrespective of presentation order. Mean Δ was 0.756 category steps (SD 1.480), median 1, interquartile range 0–2, and range -3 to 5. These are post-intervention condition differences, not changes from a pre-intervention spatial baseline; their numerical magnitudes assume comparable category steps.

Dimension	ρ_Δ	95% interval	p_Δ	ρ_{dir}	p_{dir}
Noticing	0.273	[-0.017, 0.529]	0.0872	0.286	0.0697
Not Distracting	0.097	[-0.241, 0.389]	0.5466	0.031	0.8439
Not Worrying	0.092	[-0.232, 0.415]	0.5657	0.024	0.8781
Attention Regulation	0.114	[-0.186, 0.414]	0.4809	0.005	0.9765
Emotional Awareness	-0.062	[-0.378, 0.247]	0.6992	-0.028	0.8616
Self Regulation	0.208	[-0.138, 0.501]	0.1915	0.070	0.6600
Body Listening	0.167	[-0.160, 0.459]	0.2954	0.101	0.5312
Trusting	0.376	[0.081, 0.627]	0.0160	0.335	0.0345

Table 3: Pre-experiment MAIA-2 and signed spatial change, $n = 41$. Main analyses correlate numerical Δ with each dimension; direction-only $(-1, 0, +1)$ correlations are ordinal sensitivities. All p values are unadjusted two-sided permutation values; intervals are pointwise paired-participant bootstrap intervals.

Trusting was positively associated with numerical Δ: $\rho = 0.376$, pointwise 95% CI $[0.081, 0.627]$, unadjusted permutation $p = 0.0160$ (Table 3; Fig. 8). The direction-only sensitivity was also positive ($\rho = 0.335$, $p = 0.0345$). The other seven numerical-delta correlations had $p > .05$; this does not establish absence of associations. The ancillary correlation between numerical Δ and the raw-RT Condition × Phase SOA-slope interaction

was $\rho = -0.020$, pointwise 95% CI $[-0.328, 0.294]$, $p = 0.9046$. The RT contrast is not a PPS boundary.

	MAIA only			With presentation order		
Dimension	b	p	p_{HC3}	b	p	p_{HC3}
Noticing	0.641	0.0705	0.0185	0.604	0.0962	0.0312
Not Distracting	0.387	0.1606	0.2850	0.471	0.0968	0.2179
Not Worrying	0.137	0.6371	0.7261	0.101	0.7320	0.8213
Attention Regulation	0.422	0.2313	0.2970	0.409	0.2477	0.3489
Emotional Awareness	-0.065	0.8489	0.8457	-0.079	0.8181	0.8059
Self Regulation	0.468	0.0838	0.1144	0.469	0.0843	0.1241
Body Listening	0.237	0.3573	0.3317	0.244	0.3451	0.2963
Trusting	0.613	0.0085	0.0293	0.624	0.0075	0.0235

Table 4: Condition × MAIA moderation, fitted through participant-level numerical Δ, $n = 41$ per model. Slopes are code steps per one MAIA point. Ordinary two-sided t inference is equivalent to the repeated-measures interaction $F = t^2$; HC3 provides approximate heteroskedasticity-sensitive inference. Residual degrees of freedom are 39 and 38. All p values are unadjusted; the companion exports all intervals, F statistics and model diagnostics.

The Trusting interaction slope was 0.613 code steps per MAIA point, $F(1, 39) = 7.681$, $p = 0.0085$. Its HC3 interval was $[0.065, 1.160]$, $p = 0.0293$. Accounting for presentation order gave $b = 0.624$, $F(1, 38) = 7.987$, $p = 0.0075$; HC3 95% CI $[0.089, 1.159]$, $p = 0.0235$. Noticing depended on the uncertainty estimator: ordinary $p = 0.0705$ and 0.0962, versus HC3 $p = 0.0185$ and 0.0312, without/with order respectively. This less consistent pattern is reported alongside its non-detection in the main rank analysis (Table 4).

These are exploratory associations and trait-moderation estimates, not mediation or independent replications. The analysis direction was chosen after inspecting earlier results, all intervals are pointwise, and no multiple-testing correction was applied. The 33 earlier absolute-rating, direction and behavioural associations remain available as supplementary audit outputs, alongside the nine new delta correlations and 16 moderation fits. Small sample size, uncertain reliability, category-spacing assumptions and possible confounding/carryover limit inference. Boundary-related questionnaire associations remain unestimated under the original fit gate.

3.8 Individual sigmoid proxies and plateau normalization

The additional exploratory extension obtained 168 four-cell sigmoid approximations from 42 participant records. Uniform selection among 84 candidates chose mean extraction, Gaussian local-constant smoothing with a 27-cm bandwidth, and a shape penalty of 4. Separate plateaus were primary; sharing both plateaus across each person's four curves

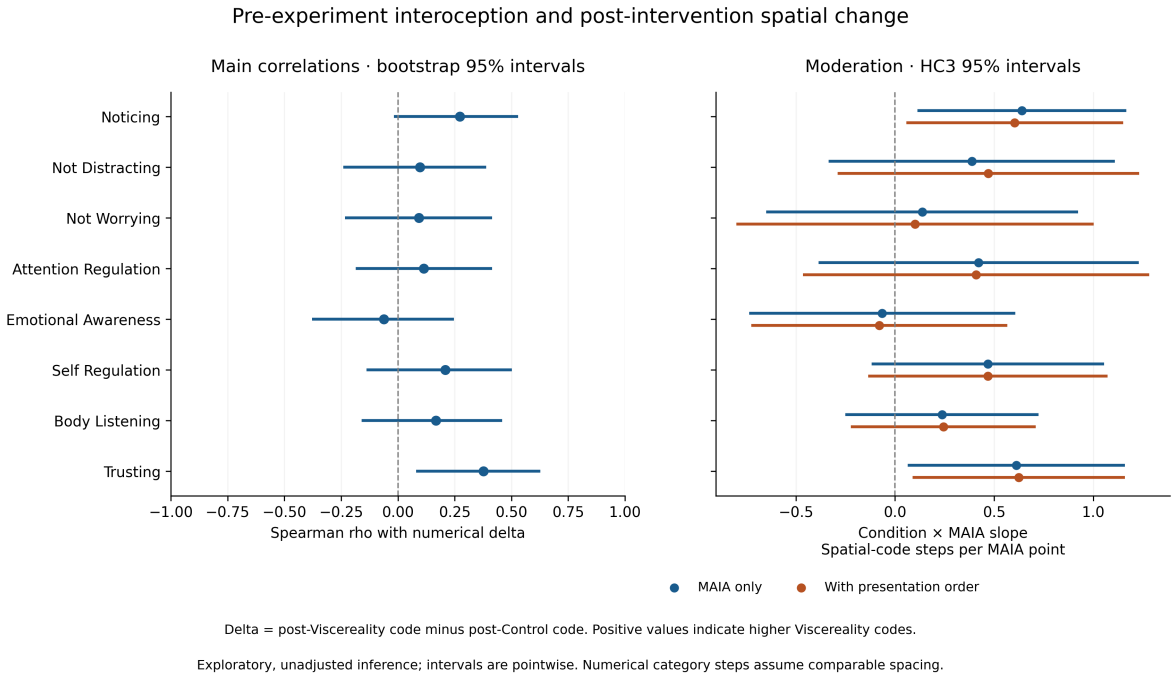


Fig. 8 Pre-experiment traits and post-intervention spatial change. Left: all eight MAIA-2 correlations with numerical Δ and pointwise bootstrap 95% intervals. Right: Condition × MAIA slopes with and without presentation order and HC3 95% intervals. Higher Δ means higher recorded Viscereality codes relative to Control; the administered pictogram direction is unverified.

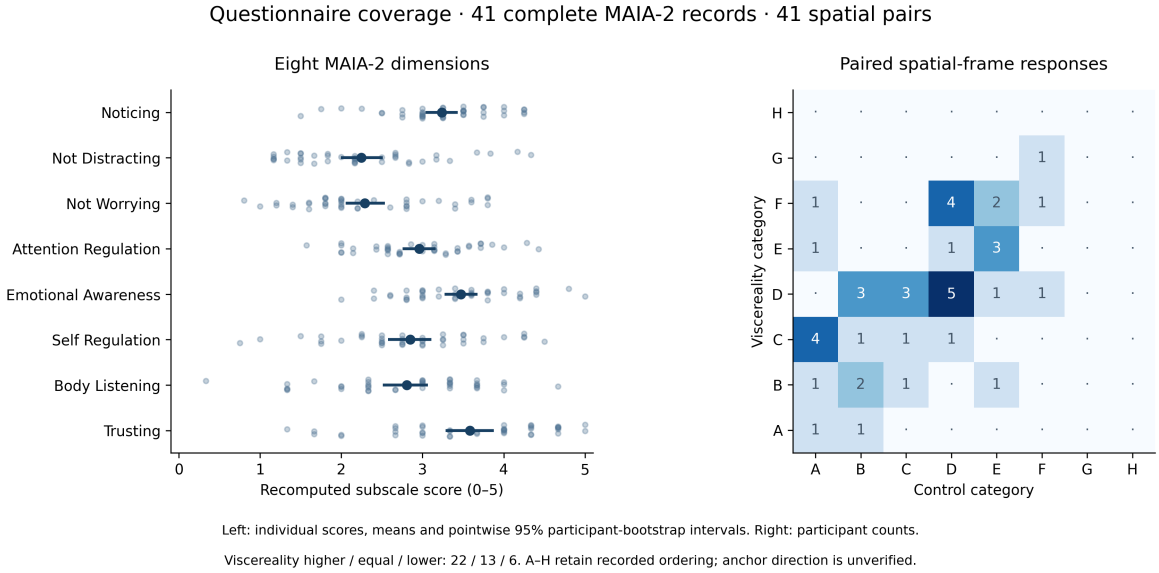


Fig. 9 Questionnaire coverage and paired spatial ratings. Left: individual recomputed MAIA-2 subscale scores, means and pointwise 95% participant-bootstrap intervals. Right: counts for all Control/Viscereality category pairs; vertical position above the matching-category diagonal indicates a higher Viscereality code. All 41 paired records are represented. A–H preserve the recorded ordering without assuming equal psychological distances or a verified anchor direction.

was a sensitivity. Expected increasing direction occurred in 104/168 primary fits; decreasing responses were retained. The visible S shape was imposed by the measured-range constraints, so numerical parameter availability does not establish physiological boundaries.

Nested omitted-position prediction RMSE was 48.507 ms for the visible family, 51.694 for broad sigmoids, 48.269 for lines, 47.437 for a constant, and 47.237 for shared plateaus (Fig. 10). The shared-plateau improvement over a constant was only 0.200 ms. These descriptive scores do not test the significance of differences between overlapping folds.

Normalizing by each fitted lower plateau and absolute amplitude gives common asymptotes 0 and 1 while preserving midpoint and width. Raw observations were not clipped. The plateau-separation bootstrap interval excluded zero for only 29/168 separate-plateau fits and 40/168 shared-plateau fits; the latter represent ten people with four repeated flags. All 42 shared-model point fits converged under the specified alternating solver, without a guarantee of global optimality. The 42-page downloadable atlas shows all four raw-RT profiles with standard errors above their normalized model geometry, with recording-linkage flags.

Midpoint split-half rank reliability was low: medians 0.047 and 0.106 for separate and shared plateaus, respectively, across ten splits and four cells. Corresponding median conditional midpoint interval widths were 39.79 and 39.00 cm. With known 80-ms sigmoids and cell-specific baselines, median midpoint recovery correlations across 30 simulations were 0.036 and 0.183. Under shared-baseline simulated truth, shared-model midpoint recovery improved to 0.360. Width-derived steepness remained poorly recovered. These checks concern the stated simulations and within-session splits; they do not measure test–retest reliability or interval coverage.

Table 5 gives all 27 main questionnaire associations for each model. Four primary or shared-model point-estimate tests had unadjusted $p < .05$, but all 54 refit-bootstrap intervals included zero. For example, the spatial-change/midpoint-change correlation moved from approximately zero under separate plateaus to $\rho = .331$ under shared plateaus. This model dependence and weak parameter reliability preclude a stable boundary association claim. Different nominal findings appeared for MAIA Not Distracting, Emotional Awareness and Attention Regulation across specifications; no finding was selected as the primary outcome after inspection. Both complete sets of 126 associations and three linkage-exclusion specifications are retained, yielding 1,008 unadjusted tests. Excluding both ambiguous records, or each separately, repeated RT-only method selection; the selected smoothing/penalty choices differed. Those sensitivity analyses cannot determine the correct recording mapping. All RT–questionnaire inference remains provisional until that linkage is resolved.

Separate plateaus			
Baseline trait / spatial change	Midpoint	Steepness	RT slope
Noticing	-0.12 (0.454)	-0.07 (0.682)	-0.01 (0.938)
Not Distracting	0.31 (0.048)	-0.15 (0.331)	-0.20 (0.202)
Not Worrying	0.25 (0.118)	0.15 (0.349)	0.05 (0.731)
Attention Regulation	-0.03 (0.830)	-0.03 (0.840)	0.17 (0.292)
Emotional Awareness	-0.34 (0.032)	-0.18 (0.254)	0.07 (0.648)
Self-Regulation	-0.07 (0.650)	-0.07 (0.647)	0.08 (0.623)
Body Listening	0.00 (0.996)	-0.28 (0.077)	-0.14 (0.367)
Trusting	0.02 (0.880)	-0.15 (0.351)	-0.00 (0.989)
Spatial code difference	-0.00 (0.994)	0.17 (0.290)	0.14 (0.375)
Shared plateaus			
Baseline trait / spatial change	Midpoint	Steepness	RT slope
Noticing	0.09 (0.560)	-0.05 (0.771)	-0.07 (0.669)
Not Distracting	0.08 (0.636)	-0.11 (0.480)	-0.22 (0.164)
Not Worrying	0.06 (0.730)	0.19 (0.220)	0.21 (0.184)
Attention Regulation	0.16 (0.311)	-0.06 (0.710)	-0.39 (0.012)
Emotional Awareness	-0.06 (0.724)	-0.14 (0.365)	-0.21 (0.181)
Self-Regulation	0.07 (0.647)	-0.08 (0.612)	-0.13 (0.419)
Body Listening	0.02 (0.907)	-0.23 (0.157)	-0.18 (0.258)
Trusting	0.30 (0.059)	-0.19 (0.237)	-0.27 (0.084)
Spatial code difference	0.33 (0.034)	0.16 (0.329)	0.17 (0.299)

Table 5: Exploratory Spearman correlations of baseline MAIA-2 or spatial-code difference with Viscereality–Control parameter differences (equal phase weights; $n = 41$). Cells give ρ (unadjusted pairing-permutation p). Parameters come from four fitted curves per person. All refit-bootstrap intervals include zero; complete intervals, cell-specific correlations and recording-exclusion sensitivities are in the companion. Full-release RT linkage is unresolved.

3.9 Recording exclusions and remaining robustness checks

An explicitly post-results amendment reran the primary behavioural models after excluding the two ambiguous records together and each separately. The numeric Condition × Phase × SOA result remained uncertain in every specification (Table 6). All three categorical-interaction and Control-only numeric/categorical comparisons also had unadjusted $p > .05$. Raw/log cluster models, equal-person slope contrasts and raw-RT repeated-measures ANOVA are retained in the companion. These exclusions test sensitivity and do not identify the correct recording-to-person mapping.

For the four fitted curves per person, equal-phase Condition, equal-condition Phase and Condition × Phase contrasts were calculated for midpoint, normalized steepness and signed RT derivative under both plateau models. All four linkage specifications received 500 person/group-plus-trial refits. Across 72 mean contrasts, 0 conditional refit intervals excluded zero; across 216 main questionnaire associations, 0 did so. For example, the

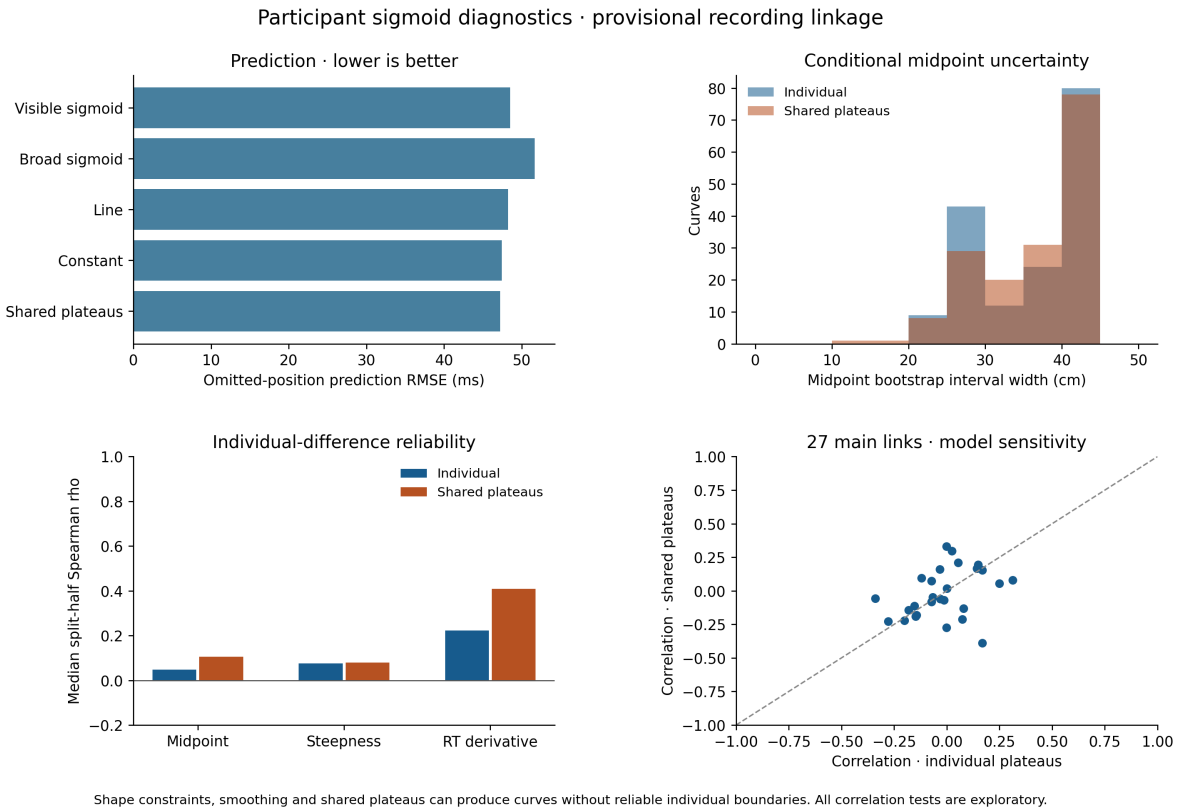


Fig. 10 Individual-parameter audit. Prediction differences are small; midpoint intervals are wide and split-half reliability is low. The lower-right panel compares the same 27 main association estimates under the two plateau assumptions. Neither normalized appearance nor numerical fit availability establishes a reliable PPS boundary.

Excluded	Records	Estimate	95% interval	p
P011, P033	40	0.86	[-9.73, 11.45]	0.873
P011	41	1.09	[-9.41, 11.59]	0.839
P033	41	0.58	[-9.92, 11.08]	0.913

Table 6: Primary numeric interaction after recording exclusions. Estimates are ms per cohort-specific SOA SD, with model-based Wald intervals; p is the unadjusted full/reduced ML likelihood-ratio comparison. Cleaning and SOA scaling were recomputed for each cohort.

separate-plateau mean midpoint change excluding both ambiguous records was -16.07 cm, with interval [-28.17, 18.16] cm. The sizeable point shift reflects a fitted, partially pooled proxy; the wide interval and weak reliability prevent interpreting it as demonstrated PPS contraction.

The changes used in correlations were themselves unstable. Across ten splits, median rank correlations for midpoint change were -0.074 and 0.066 for separate and shared plateaus, respectively. Corresponding normalized-steepness values were 0.052 and -0.009; signed RT-slope values were 0.064 and 0.155. Known-signal simulations also assessed all three contrasts. Shared-model midpoint-change recovery was better when truth shared plateaus,

but this simulation-specific improvement does not establish reliable observed individual differences (Fig. 11).

The group-response audit excluded both ambiguous recordings ($n = 40$) and added 160 log-distance pipelines with the same extraction/smoothing choices as lines. Six-outer/five-inner participant/position prediction gave RMSEs 47.993 ms for broad sigmoids, 47.915 for visible sigmoids, 48.087 for lines, 48.121 for log distance and 48.020 for a constant. The visible family's improvement over a constant was 0.105 ms; this small descriptive difference is not a significance test or physiological validation. The log-distance comparison does not reproduce the expectancy-controlled design of Hobeika et al. (2020). Unsmoothed nonlinear-component tests in the four curves all had unadjusted $p > .05$.

Nine fixed geometry specifications crossed three width floors with permissive, existing and strict visibility criteria. Their descriptive leave-position RMSE ranged from 47.47 to 94.59 ms; the large error under one permissive specification illustrates instability when an omitted position leaves the transition poorly constrained. Thirty whole-person resamples repeated all 320 visible-method selections with repeated copies kept in the same validation fold. They selected 27 distinct pipelines, and the most frequent appeared only 3 times. These finite sensitivity checks retain all outcomes; thirty draws describe selection instability and do not provide calibrated selection-adjusted tail intervals.

3.10 Reproducibility and evidence limits

All reported behavioral computations were executed in the marimo browser companion with Pyodide, Python 3.14.2, NumPy 2.4.3, SciPy 1.17.1, and Matplotlib 3.10.8. The archived browser download records input, code, and output SHA-256 hashes. Native reference checks independently verified random-intercept ML fits, cluster-robust linear and binomial coefficients/uncertainty, repeated-measures F statistics, and JZS quadrature. Additional native references checked questionnaire alpha through its covariance identity, the exact ordinal sign test and interval, all 42 rank correlations, and all 16 moderation fits including ordinary and HC3 uncertainty. Independent long-format participant-intercept models verified the delta regression's Condition $\times$ MAIA slope, standard error, F statistic and p value. Every questionnaire result was also compared across native and browser runtimes. For individual proxies, independent KernelReg smoothing and scalar least-squares grid objectives checked all 168 primary fits; weighted least squares checked shared plateaus conditionally, and SciPy independently checked correlations across both models and recording exclusions. Every individual result was compared across native and browser runtimes. A failed or unavailable check is distinguished from a computed null result.

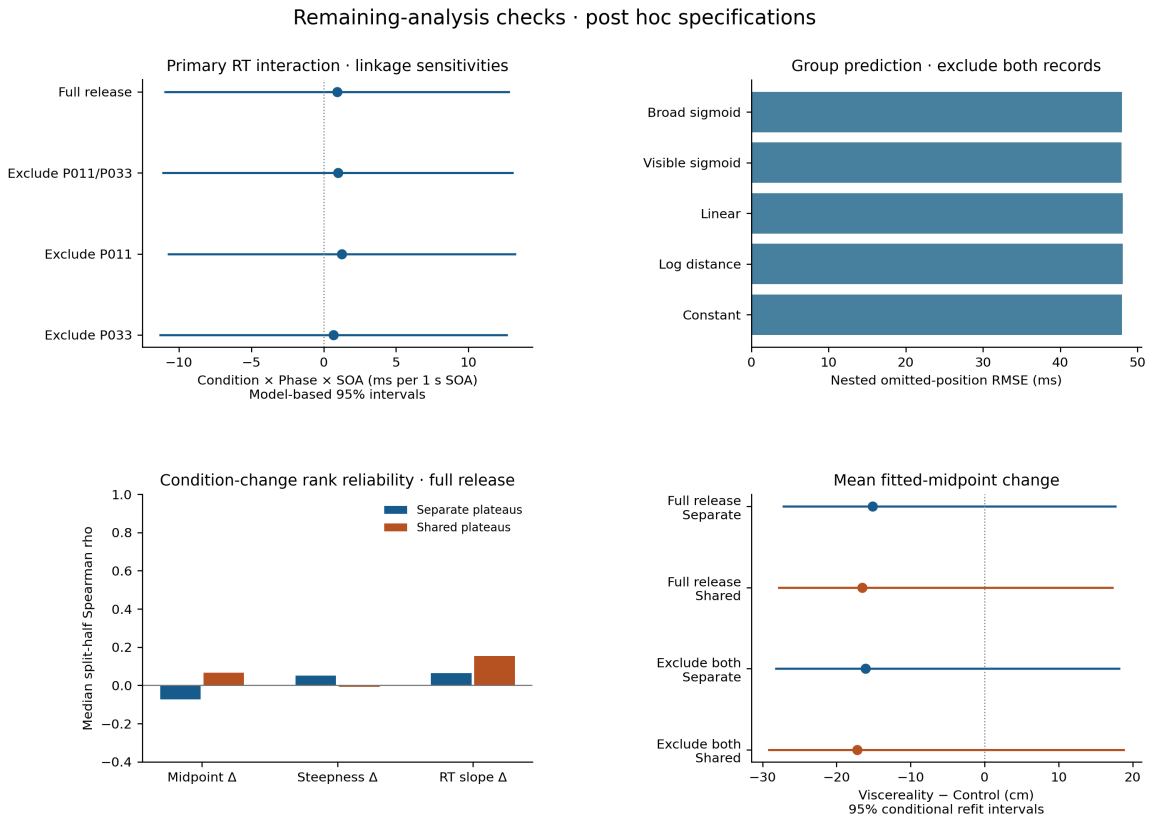


Fig. 11 Post-results robustness extension. Top left: primary RT interaction and model-based uncertainty across recording exclusions, rescaled to a common one-second SOA increment. Top right: five response families on the same 40-record prediction target. Bottom left: reliability of fitted Condition differences. Bottom right: midpoint-change point estimates and intervals that refit the models. Large fitted shifts can coexist with very uncertain, unreliable changes.

The released CSVs do not contain waveforms, belt recordings, acquisition-version labels, collection dates, or the private provenance needed to identify two historically documented RT-onset variants. The browser analysis therefore does not rerun waveform timing, measured respiratory compliance, pilot-date classification, version-specific contrasts, or that source-definition exclusion. Historical source audits remain historical evidence, rather than new 42-person verification.

4 Discussion

The full-release analysis did not detect the predicted Condition × instructed Phase × SOA interaction. Its inference is provisional: two of the 42 participant records contain identical behavioural recordings with reversed intervention labels and different questionnaire responses. Resolving that source mapping is necessary before finalizing the behavioural cohort and its links to questionnaires. The unchanged full-release calculations document

the current data, not 42 verified independent recordings. Numeric and categorical coding, partial-session exclusion, participant-cluster-robust uncertainty, and equal-participant slopes led to the same qualitative conclusion, but none resolves this linkage issue. The intervals permit effects in either direction, so the result does not establish equivalence or the absence of respiratory influences on PPS.

Audio–Tactile responses were faster than tactile-only responses on average, but neither the raw SOA gradient nor subtraction of Baseline at the same nominal delay label isolates auditory proximity. Historical source checks found unequal trial-relative tactile-cue times for equal labels; the public tables cannot rerun those waveform checks. Temporal expectancy, response probability, trial counts, and control conditions matter when interpreting near-versus-far tactile benefits (Holmes et al., 2020). Baseline phase contrasts were uncertain after adjustment across the supplementary family; these data establish neither phase invariance nor an adjusted Baseline phase effect.

The participant-level sigmoid gate failed in every trial cell. The saturated five-SOA ceiling shows that the required trial-level variance explanation is unattainable for any deterministic curve of these SOAs, independently of nonlinear optimizer settings. Better in-sample sigmoid than linear RMSE therefore does not supply valid participant-boundary estimates. Four parameters fitted to five SOA means leave one residual degree of freedom and often meet parameter limits; group curves likewise cannot provide participant replicates. The disagreement between approximate registered distances and the stated trajectory changes some descriptive fit diagnostics but leaves the trial-level gate failure unchanged. Participant-boundary, sharpness, and related questionnaire hypotheses consequently remain unestimated under the planned gate.

The separate descriptive extension obtained four sigmoid approximations through a uniform signal-extraction and smoothing procedure. The original 320-pipeline winner yielded broad transitions whose 10–90% spans all exceeded the measured range. Requiring a visibly S-shaped transition produced four sharper approximations, but this geometry was imposed: three widths reached their floor and one midpoint reached the sampling limit. In nested participant prediction, the constrained family slightly improved on equally processed lines but not on a constant. Unsmoothed curvature tests likewise did not establish nonlinearity. Neither result proves that the population response is linear or constant; together they leave a data-supported PPS sigmoid unresolved. The midpoint and slope comparisons are therefore assumption-dependent descriptions. Conditional bootstrap intervals omit method-selection uncertainty and are restricted by the imposed geometry. A future design with more positions spanning both plateaus, suitable expectancy controls,

and an independently specified extraction/bandwidth rule would allow a stronger test of midpoint and sharpness differences. The later 40-record exclusion audit compared that logarithmic alternative under the same nested target. The visible sigmoid narrowly outperformed the constant by 0.105 ms, while the log-distance family did not improve prediction. This small descriptive difference does not establish an S-shaped mechanism. Nine prespecified-for-this-extension geometry specifications exposed substantial sensitivity under a permissive rule, and 30 repeated selections chose 27 distinct pipelines. These findings strengthen the need to separate numerical approximation from identified physiological parameters. They do not replicate the expectancy-controlled experiment of Hobeika et al. (2020), calibrate method-selection-adjusted intervals or exhaust plausible shape definitions.

The individual-parameter extension achieved a computational description for all four cells of every participant record, including normalized common plateaus. It did not establish reliable individual boundaries. Shared plateaus slightly improved prediction and some recovery measures, but midpoint split-half reliability remained low, uncertainty was wide and the main questionnaire links changed with plateau assumptions. Every main refit-bootstrap interval included zero. The penalty can stabilize numerical fitting while shrinking genuine differences; unit-amplitude normalization can make negligible or uncertain changes look prominent. The raw-RT panels, direction flags and amplitude intervals therefore remain necessary companions to the normalized S shapes. Partial pooling and recovery studies provide a methodological rationale (Ahn et al., 2011), while the distinction between group effects and individual-difference reliability (Hedge et al., 2018) explains why attractive curves alone cannot support correlations. The three recording-exclusion analyses now include the primary RT models and refit intervals for parameter contrasts and questionnaire links. The primary RT conclusion was unchanged; all 72 mean-contrast and all 216 main association intervals across cohorts and plateau models included zero. Midpoint-change split-half correlations were approximately zero, directly limiting the use of those changes in individual-difference correlations. These analyses expose sensitivity but do not recover the missing source mapping. These post hoc proxies cannot substitute for the failed preregistered boundary analysis.

The questionnaire extension adds a post-intervention experiential outcome alongside the behavioural findings. Paired spatial-frame codes tended to be higher under Viscereality, despite both condition medians being D. The owner confirmed that MAIA-2 preceded the experiment, supporting its use as a baseline trait correlate/moderator. Higher Trusting was associated with a larger signed Viscereality–Control code difference. Its positive association appeared in numerical-delta and direction-only correlations and in Condition × MAIA models with ordinary and HC3 uncertainty, with and without presentation order. These

analyses reuse the same participants and do not constitute independent replications. Noticing was less consistent: HC3 interaction inference differed from the ordinary interaction and main rank analyses. The other dimensions did not show clear numerical-delta associations, which does not establish independence. Numerical differences assume comparable category spacing; the direction-only sensitivity avoids that spacing assumption. Without the administered pictogram key, higher codes cannot yet be assigned an allocentric or egocentric interpretation. Order adjustment does not remove all confounding or resolve carryover. These post hoc, unadjusted findings describe associations and trait moderation, not mediation or a PPS-boundary mechanism. MAIA-2 self-report is conceptually distinct from objective interoceptive accuracy (Garfinkel et al., 2015; Mehling et al., 2018). The small sample, uncertainty in reliability and multiple comparisons limit inference. Original questionnaire registration wording, administered anchors and language still require verification before submission.

Several features constrain physiological interpretation. Phase labels identify the instructed half-cycle rather than measured, trial-by-trial belt phase. Historical acquisition sources differ in RT definitions, available sounds, and recording availability; the released CSVs do not identify versions or support uniform waveform harmonisation. The hit-probability and time-on-task checks do not identify RTs on missed trials. The missing-hit-RT scenarios probe specified departures from cell medians and cannot establish the missingness mechanism. The recorded behavioral results should therefore not be read as a precise test of physiologically measured PPS boundaries.

Registration distinctions also require transparent reporting. The Bayesian mixed-model comparison was not fully specified, so the exploratory paired JZS factor addresses a different model and is sensitive to its normal-likelihood assumption. The sigmoid exclusion's scope and computational details required explicit interpretation. The registration disclosed pilot observations, and the released roster exceeds its stopping target; the collection history and ethics-statement details require author documentation before submission. Those items cannot be inferred from arbitrary public participant IDs.

The marimo publication companion makes the data-supported analysis inspectable and executable in the reader's browser. It exposes preprocessing counts, fixed analysis choices, model uncertainty, failed fit gates, unavailable provenance checks, and downloadable source and result hashes. Independent native checks support numerical agreement without substituting precomputed results for browser execution. This reproducibility supports assessment of the reported null detection while preserving the distinction between an executable analysis draft and a paper ready for submission.

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